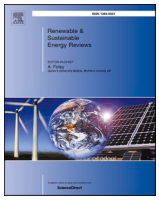
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A comprehensive review on optimization of hybrid renewable energy systems using various optimization techniques

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ABSTRACT

The increasing energy prices and pollutants from fossil fuels that threaten the climate, there is a growing preference for renewable energy. The implementation of hybrid renewable energy systems (HRES) has been a challenging task due to its interference, uncertainty, and unpredictable nature. Also, it comes with high net present cost and multi-dimensional architecture facets. It is critical to evaluate HRES using a variety of economic and sizing criteria. This paper aims to provide a succinct review of recent progress in the field of optimization of different HRES using various optimization techniques based on classical methods, artificial intelligence (AI), hybrid algorithms, and software-based optimization tools. While comparing classical and AI-based techniques, the AI-based techniques are found promising and provide a global solution in less time. However, there are deficiencies in AI-based techniques. To overcome those, a combination of two or more algorithms, known as hybrid optimization algorithm can be adopted to solve problems more quickly, reliably, and effectively. Apart from different algorithms, there are several popular optimization software tools. Among them, the HOMER software tool is one of the most popular as it is simple to use. As the research activities in the optimization of HRES are increasing, the concentration on sources such as hydro, geothermal, biomass, and biofuel has to be given due consideration. This analysis will inform readers about the current and evolving state of optimization approaches for HRES applications and enable them to choose the most appropriate strategy as per requirement.

1. Introduction

Nowadays, the world is facing a great challenge in determining the solution to exponentially increasing energy demand and the problems associated with fossil fuel resources. Through the expansion of economies and the global population, the demand for energy rises, resulting in a rise in fossil fuel use. Current conventional resources, such as coal, petroleum, natural gas, and so on, are not environmentally friendly, are expensive, and pose a negative threat to future power demand and generation balances. However, conventional fuel supplies are scarce and quickly depleting, necessitating intensive attention and sustainable interventions to avert future energy crises. Additionally, fossil fuels contribute toxic emissions, such as greenhouse gases, to global warming. Renewable energy is being applied along with non-renewable sources to supplement the primary energy demands and is particularly used to optimize energy efficiency. Instead of relying on a single type of power source, hybrid technology is the most popular choice for lowering costs and increasing system reliability. The key issue in sustainable energy advancement is the reliability aspect and mathematical optimization of hybrid distributed energy networks for operation and planning [1]. Renewable energy systems designed for a specific application can be considered to be too large or excessively small. This exacerbates the problem because the unpredictability of the energy supply will not be sufficient to satisfy demand under such circumstances. The energy generated by renewable energy systems has risen in recent years as a result of advancements in technology and through the increased arousal of the populace worldwide. In India, the sizable majority of people reside in rural and remote locations that are disconnected from the grid, and expanding the grid is prohibitively expensive. As a result, those regions can be electrified in a decentralized way using renewable energy sources such as small hydro plants, biomass, PV, and wind, as well as their integration [2]. However, renewable energy sources are intrinsically stochastic, which is their primary disadvantage. Unlike fossil fuel-based energy production which is predictable and entirely controllable, renewable energy-based electricity production is inherently unreliable, and it is inextricably connected to the atmosphere, making it impossible to control with the accuracy required for a reliable

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Nomenclature

ABC Artificial Bee colony
ABSO Artificial bee swarm optimization
ACO Ant colony optimization
ACOR Ant colony optimization for continuous domains-based integer programming
AKQPSO Hybrid Annealing Krill Herd and Quantum-Behaved Particle Swarm Optimization
ALC Annual Levelized Cost
ALO Ant lion optimizer
ANN Artificial neural network
AP Acidification Potential
APSO Accelerated Particle Swarm Optimization
ARC Absorption refrigeration cycle
B&B Branch and bound algorithm
BA Bat algorithm
BAPSO Hybrid bat-particle swarm optimization
BBKH biogeography-based KH
BBO Biogeography-based optimization
BES Battery energy storage
BESS Battery energy storage system
BFSO Butterfly particle swarm optimization
BPFPA Hybrid Bee pollinator Flower Pollination Algorithm
BSA Backtracking Search Optimization Algorithm
BWO Black Widow Algorithm
CCS Chaotic cuckoo search
CED Cumulative Energy Demand
CHP Combined heat and power
CS Cuckoo search
CSA Crow search algorithm
DE differential evolution
DEKH Differential evolution KH
DHA Deer hunting algorithm
DHS discrete harmony search
DMGWO Discrete Multiobjective Grey Wolf Algorithm
DPP Deficit power probability
DPSP Deficiency of Power Supply Probability
DSA discrete simulated annealing
EA Evolutionary algorithms
EDE Energy demand estimation
EEF Excess energy fraction
EENS Expected energy not supplied
EIR Energy index of reliability
EL Electrolyzer
ELF Equivalent loss factor
ELM Extreme Learning Machine
EMD Empirical Mode Decomposition
ENBIOTEM ENergy from BIOmass Techno-Economic Model
EUE Expected Unserved Energy
EV Electric vehicle
FA Firefly algorithm
FASSO Fuzzy adaptive swallow swarm optimization algorithm
FBHPSO-DE Fuzzy-Based Hybrid Particle Swarm Optimization-Differential Evolution
FC Fuel cell
FEEMD Fast Ensemble Empirical Mode Decomposition
FEVCS-WPE Fast EV charging stations with wind, PV power, and energy storage system
FPA Flower Pollination Algorithm
FSO Fluid Search Optimization
FSOWCO Hybrid Flower Pollination Algorithm and Fluid Search Optimization
GA Genetic algorithm
GC Grid connected
GEP Global Warming Potential
GEWA Generalized Evolutionary Walk Algorithm
GFM-MOEA Generic front modeling-based multiobjective evolutionary algorithm
GPAP Grid power absorption probability
GPSBBO Greedy particle swarm and biogeography-based optimization
GSA Gravitational search algorithm
GSA–TLBO Hybris Gravitational search algorithm and teaching learning-based optimization
GT Gas turbine
G-TLBO Genetic Teaching Learning-Based Optimization
GWO Grey wolf optimizer
GWOCS Grey wolf optimizer and cuckoo search
HBB-BC Hybrid Big Bang–Big Crunch
HBSO Henry gas solubility optimization
HCHSA Hybrid chaotic search/harmony search/simulated annealing
HDI Human development index
HFA/HS Hybrid Fire-Fly/Harmony Search Optimization
HF High frequency
HGSO Henry gas solubility optimizer
HHO Haris hawks optimization
HIP Hourly interruption probability
HIS-IHS improved harmony search
HIWO–PSO Hybrid Invasive Weed optimization - Particle Swarm optimization
HOMER Hybrid optimization model for electric renewables
HRES Hybrid renewable energy sources
HRSG Heat recovery steam generator
HS Harmony search
HS-GIS Harmony search-geographic information system
HS-SA Harmony search and simulated annealing
HSWPS Hybrid solar-wind power system
HT Hydrogen tank
Hybrid HS-SAA Hybrid harmony search-simulated annealing Algorithm
IACA Improved Ant Colony Optimization
IAEO Improved artificial ecosystem Optimization
IA-PSO Immune algorithm-based particle swarm optimization
IBA Improved bee algorithm
ICA Imperial competitive algorithm
ICSA Improved crow search algorithm
IHSBSA Improved harmony search-based simulated annealing
IHS-SA-GIS Improved harmony search, simulated annealing, and a geographic information system
IPSO Improved particle swarm optimization
ISGA improved gravitational search algorithm
ITSA Improved Tunicate swarm algorithm
IWO Invasive Weed Optimization
IWPSO Inertia weight particle swarm optimization
JC Job creation
KH Krill herd algorithm
LCOE Levelized cost of electricity
LDR Load Dissatisfaction Rate
LF Low frequency
LOEE Loss of energy expected
LOLE Loss of load expected
LOLE Loss of load expected
LPSP Loss of power supply probability
LQE linear, quadratic, and exponential
LUEC Levelised Unit Electricity Cost
MBA Mine blast algorithm
MCS Monte Carlo simulation
MDPSO Multimodal Delayed PSO Algorithm

MEA	Mind Evolutionary Algorithm
MFABC	Multi-strategy fusion artificial bee colony
MFFA	Modified Farmland Fertility Optimization Algorithm
MFO	Moth–flame optimization algorithm
MILP	Mixed integer linear programming
MLP	Multi-Layer Perceptron
MLUCA	multi-objective line-up competition algorithm
MOCSA	Multi-objective crow search algorithm
MOEA	multi-objective evolutionary algorithm
MOHTLBOGWO	Multi-Objective Hybrid Teaching–Learning Based Optimization-Grey Wolf Optimizer
MOOD	Multi-objective optimization dispatch
MOPSO	Multi-objective particle swarm optimization
MOSaDE	multi-objective self-adaptive differential evolution algorithm
MSFLA	Modified shuffled frog leaping algorithm
NBA	Novel bat algorithm
NSWOA	Non-dominated sorting whale optimization algorithm
OEHO	Oppositional Elephant Herding Optimization
ORC	Organic Rankine cycle
PC	Penalty cost
PEMFC	Proton exchange membrane fuel cell
PIO	Pigeon-inspired optimization
PO	Political Optimizer
PSO	Particle swarm optimization
PSOGSA	Particle swarm optimization and gravitational search algorithm
PSOMCS	Particle Swarm Optimization Algorithm-based Monte Carlo Simulation
PSOSCALF	PSO based on Sine CosineAlgorithm and Levy flight
PVSYT	Photovoltaic (PV) System (syst)
QKH	Quantum-behaved kill herd
QPSO	Quantum-Behaved Particle Swarm Optimization
RBFNN	Radial Basis Function Neural Network
RBFNOEHO	Radial Basis Function Neural Network and Oppositional Elephant Herding Optimization
RC	Rankine cycle
RCGA	Real-coded genetic algorithm
RCSOMGA	Real-coded self-organizing migrating genetic algorithm
RDS	Radial distribution system
RF	Renewable fraction
RO	Reverse osmosis
SA	Simulated Annealing
SA-GIS	Simulated annealing, and a geographic information system
SAPSO	Simulated annealing particle swarm optimization
SFLA	Shuffled frog leaping algorithm
SFO	Sailfish optimizer
SKH	stud krill herd
SOA-TEO1	Hybrid seagull optimization algorithm and thermal exchange optimization1
SOA-TEO2	Hybrid seagull optimization algorithm and thermal exchange optimization2
SOA-TEO3	Hybrid seagull optimization algorithm and thermal exchange optimization3
SOMGA	Self-organizing migrating genetic algorithm
SOP	Single objective problem
SPPA	Strawberry Plant Propagation Algorithm
SSO	Sparrow search optimization
SVM-BAT	Space vector BAT optimization
SVM-PSO	Space vector machine-particle swarm optimization
SVM-WOA	Space vector machine-Whale optimization algorithm
TAC	Total annual cost
TLBO	Teaching-learning-based optimization
TLBO-CMMI	Teaching–learning-based optimization employing correction matrix method
TSA	Tunicate swarm algorithm
VNBA	Variable neighborhood search to bat algorithm
WCA	Water cycle algorithm
WCO	World cup optimization
WHR	Waste heat recovery
WOA	Whale optimization algorithm
WPD	Wavelet Packet Decomposition
WSA	Water Strider algorithm
WSH	Wind solar hydro
WTE	Waste to energy

electricity supply. To address this problem, hybrid renewable energy systems that combine two or more energy sources can compensate for one source's shortcomings with the strengths of another. For instance, regular and seasonal patterns of wind and solar irradiation complement one another. Additionally, HRES can be integrated with a more predictable energy source, such as on-demand biomass fuel [3]. A hydropower facility, like a massive energy storage plant, has a high degree of adjustability. The power output of a hybrid system with a hydropower station can be easily smoothed [4]. Energy storage technologies types such as batteries, fuel cells, flywheels, electrochemical/super/ultracapacitors, molten salt, compressed air energy and pumped hydro storage can be a viable option for addressing the issues associated with renewable energy's erratic existence [5]. HRES models are off-grid for remote areas where conventional fossil fuel generation and grid extension are not feasible, and HRES can also operate in a grid-based mode. As a result, HRES capabilities contribute to global protection, economic efficiency, quality, dependability, and the potential for increased social indexes. Since these resources are entirely location-specific, finding a satisfactory blend that combines both power supply efficiency and economics is a challenging one. The graphical abstract of the article is presented in Fig. 1.

Hybrid system analysis is highly difficult and must be extensively examined due to the various generating systems. The optimization goals of renewable-based hybrid systems are to achieve optimal operating conditions and economic evaluation and optimize their performance to the point where all their physical and technical constraints are satisfied. The goal of sizing a hybrid renewable energy system serves as a reminder to use renewable energy sources effectively and affordably. It is essential to maximize the power on sunny or windy days with the optimum designed design since the electricity provided by renewable energy sources like wind and solar is mostly dependent on meteorological conditions. The efficient use of renewable energy sources depends on the techno-economic analysis of the hybrid system. The application of various optimal sizing techniques in the design of hybrid energy systems is becoming more and more popular. Because the hybrid system's optimal configuration and control method are interconnected, it is more challenging to size, design, and evaluate hybrid systems. Providing an optimal sizing approach can aid in ensuring the highest performance and the lowest investment while making the most use of the system component. As the number of variables and parameters associated with hybrid renewable energy systems increases, optimization techniques are required to effectively size the system components. To obtain the optimal combination of components size of the system, the most economical and reliable system the sizing and optimization techniques require an effective search to avoid under as well as oversizing of the hybrid energy system. Flexibility, resilience, and sophisticated computational intelligence in optimization methods can tackle challenging problems in hybrid renewable-based systems. As the hybrid renewable energy system is one of the most trustworthy alternatives to mitigate climatic change and minimize the use of fossil fuels, it is

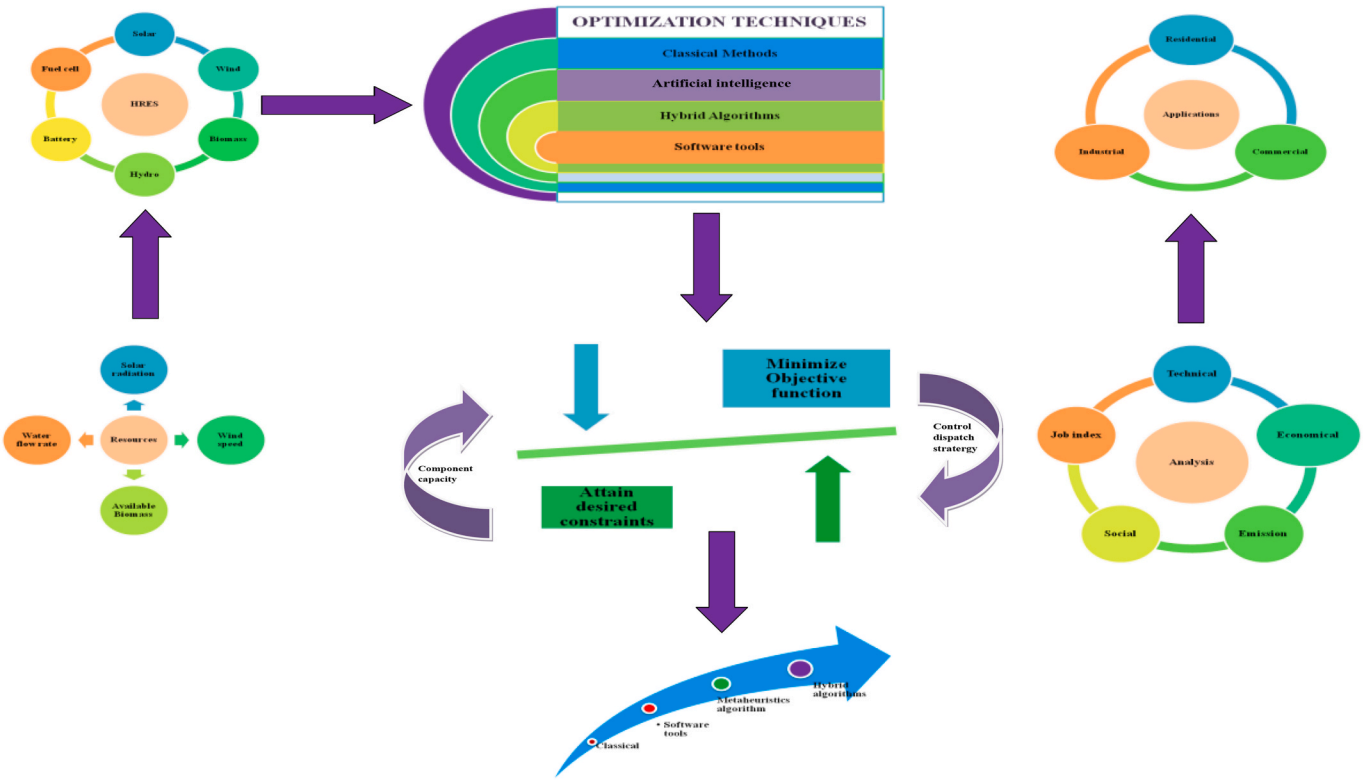


Fig. 1. Graphical abstract of the article.

necessary to design and optimize the system components effectively to improve the system's reliability and reduce investment costs. Thus, to help the stakeholder and researchers carry out their further research in the field of optimal design of energy systems, an updated literature review about the latest state of art of hybrid renewable energy systems, their applications, and an extensive analysis of the most recently implemented optimization methods and techniques is required. In general, optimization algorithms can be classified as classical approaches, artificial intelligence, Hybrid techniques, and software-based optimization tools.

In the past, some authors have reviewed systems based on solar-PV and wind systems [6], Systems based on solar energy [7], a combination of both PV and wind-based energy systems [8], Computational tools, and some artificial intelligence techniques for sizing methodologies [2], Optimization technique applied to undertakes horizontal wind turbine performance optimization [9], Various modeling and optimization techniques employed for the application of off-grid based HRES [10], Various configuration and control strategy of HRES [11], Operation and planning of HRES [12], Main features of nineteen optimizing software tools [13], Review on HOMER software for optimal sizing of HRES [14]. Looking through all of these review studies, all presented an optimal sizing system based on either solar or wind only, but none provided a detailed review of the economic, environmental, objectives, reliability, social, and technical bases of HRES modeling, comparison of optimal results obtained from various techniques, a detailed review on hybrid systems based on biomass and hydro energy systems, a review on the application of the recent evolution of artificial intelligence techniques, including hybrid techniques used for sizing hybrid HRES.

This paper presents a review of the latest research published on the optimal sizing of hybrid renewable energy systems. Unlike other existing reviews, a brief review of the optimization techniques that have all recently been employed to optimize HRES has been carried out, without limiting it to any particular hybrid system, a particular issue, a particular application, or a particular geographic site. The various objectives, decision variables, and reliability constraints used to define the optimization problem are summarized. This review article informs readers about recent advances in the field of techno-economic feasibility analysis of hybrid renewable energy systems and serves as a platform for future research in the area. Also for readers, this paper can be used to choose the best combination of renewable energy sources for a specific location to meet that location's supply demand. Section 2 presents an overview of types of optimization problems and optimization techniques. Section 3 discusses the different types of classical optimization methods applied to optimize HRES. Section 4 summarizes Artificial intelligence techniques applied to optimize HRES. Section 5 reviews hybrid algorithms applied to optimize HRES. Section 6 talks about the recently developed artificial intelligence algorithms and their applications in various fields. Section 7 presents a review of the software tools related to optimization. Section 8 highlights the discussions and findings from the study. Section 9 concludes the paper.

2. Overview of types of optimization problems and optimization techniques

The task of deciding the best approach for minimizing or maximizing a problem's objective function is known as optimization [15].To solve engineering problems under given circumstances and constraints, optimization processes have been introduced for different applications. For a given problem, mathematical optimization can be defined as the process of obtaining the best strategy or determining the conditions to maximize desired factors or minimizes undesired factors, termed an objective function. A mathematical interrelationship between the objectives, constraints, and decision variables attempts to determine the appropriate algorithm and shows how hard it is to obtain the optimal solution for an optimization problem. The types of optimization problems are classified in terms of the number of objectives, types of variables, types of constraints, nature of optimization, nature of the equation, and problem structure. In recent years, a large number of control variables have had to be optimized according to a range of constraints in a variety of real-world optimization problems. The design optimization is

evaluated based on the number of objective parameters as a single or multi-objective approach. A single-objective approach deals with only one objective function but does not satisfy several sets of objectives simultaneously. It is difficult to maintain proper diversity in evolutionary algorithms for single-objective functions; it leads to premature convergence due to loss of diversity. A multi-objective approach will simultaneously treat several numbers of objectives. To optimize the objective function, the appropriate assignment of optimization variables, which will be varied within their upper and lower bounds in the optimization expression, is necessary. Fig. 2 represents the types of optimization problems.

To achieve the optimization goal, optimization techniques are developed, which are generally classified into classical techniques, artificial intelligences, and software tools. Classical methods approach differential calculus; artificial intelligence approaches brain learning and training processes without requiring any mathematical modeling; Artificial intelligence techniques are a higher-level procedure, and software tools are based on a set of computer programs. Fig. 3 represents the general classification of optimization techniques. A list of a few optimization techniques and their inspiration, merits, and demerits is provided in Table 1. While designing a hybrid renewable energy system selection of evaluation parameters such as technical, economical, reliable, environmental, and social should be necessary to analyze the feasibility of the optimal design. The components of the hybrid energy system are very expensive; determining the optimal sizing of individual components in a hybrid energy system would result in lower system life cycle cost, net present cost, electricity cost, and cumulative cost savings over the life of the system. Reliability is defined as the probability of the system performing effectively over the operating time. It is also defined as the power system's ability to provide a reliable, adequate, continuous, secure, and quality supply to the load. To improve the behavior of system performance the evaluation of reliability indicators such as loss of load, loss of energy, loss of power supply, etc is necessary. A summary of the commonly employed cost, reliability, social, and environmental indices, and their main formulas are provided in Table 2.

3. Classical optimization methods applied to optimal sizing of HRES

Optimization based on classical techniques is deterministic, using mathematical formulations to determine the global optimal solutions. The fact that it is capable of providing a definite answer is one of its merits, but it can be observed that it is unable to handle a large number of variables in a complex space. In the early stages of design and optimization, classical techniques typically used Numerical, Iterative, Analytical, and probability analysis, as well as graphical [3]. These methods for deriving the solution are all derived using differential calculus to calculate the most efficient energy models. Classical dependent optimization techniques make it easier to find optimal continuous and differentiated solutions. When the challenges include objective functions that are not continuous and/or differentiable, these optimization methods have a limited scope.

3.1. Numerical methods applied to optimal sizing of HRES

Numerical methods are techniques that determine approximate rather than exact solutions to problems by using mathematical tools or mathematical procedures. An off-grid solar/wind/diesel/battery hybrid generation system was designed, and a techno-economic optimization based on reliability and an economic model was carried out for a site in Ghardaia, Algeria. An iterative optimization approach has been employed to predict the optimal configuration based on the minimum cost. A reliability model based on total energy deficit and an economic model based on net present cost and electricity cost has been used to perform techno-economic analysis to attain zero energy deficits. An 8.5 kW PV system, a 1 kW wind turbine, a 4.2 kVA generator, and an 86.4 kWh battery are the optimal configuration for a solar/wind/diesel/battery hybrid generation system [54]. An approach to the optimal configuration of an off-grid hybrid wind/solar PV/hydrogen system to supply a desalination unit has been proposed. This approach is based on an iterative technique to optimize the size of solar PV units, wind generators, fuel cells, electrolyzers, the capacity of the hydrogen tank, and the size of the desalination unit system so that the total system cost and the hybrid system's energy storage needs are kept low, thus ensuring a stable power supply. From the obtained results, it can be observed that due to the good penetration of solar and wind sources in the Kerkennah Islands, the required capacity of the hydrogen tank has been reduced, which improves the system reliability [55]. Both deterministic and stochastic approaches are proposed to determine optimal solar PV power for the residential PV micro-systems using net metering. This analysis shows stochastic optimizations reach an optimal power of 18.93 KW, which is lower than the deterministic approach [56]. The determinist approach is used to optimize the size of generation and storage units to curtail the annual capital cost and cost of electrical energy of HERS, to supply for an urban residential application considering three different cases of load consumption in the region of Monastir, Tunisia. Three different distributions of the residential demand profile are considered, and the results show that a PV/wind/diesel hybrid energy system with electrochemical storage is economical, where the cost of electricity obtained lies between 0.3082 and 0.3716 € per kW-h [57]. A Stochastic approach of optimization is suggested to find the ideal size of an energy storage system under wind uncertainties to enhance micro-grid accuracy and reliability. Integrating the optimal capacity of batteries with micro-grid based on renewable sources reduces the occurrence and duration of outages [58]. A discrete tabu search approach for optimizing three different designs of standalone systems comprising solar, wind, and battery systems is proposed for various regions of Iran under different operating conditions. Based on the system's annual cost, the obtained results are compared with the harmony search. The results of the proposed algorithm show better performance and effectiveness than the compared one. Based on the economic analysis, the PV/Battery combination is the first preference for the selected location due to high solar radiation, and PV/WT/battery may be an appropriate choice [59]. Mixed integer linear programming is applied to evaluate the performance of a standalone hybrid solar PV/wind/battery system to meet South-Tyrol mountain region demand. The technical and economic analysis of the system was chosen with the objective of optimizing system performance. As the initial cost of the battery is high, the total

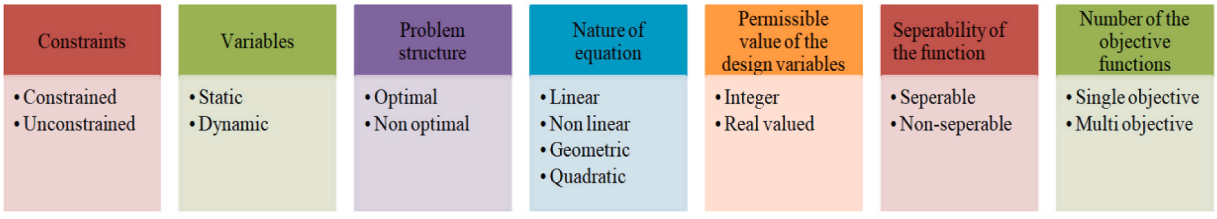


Fig. 2. Types of the optimization problem.

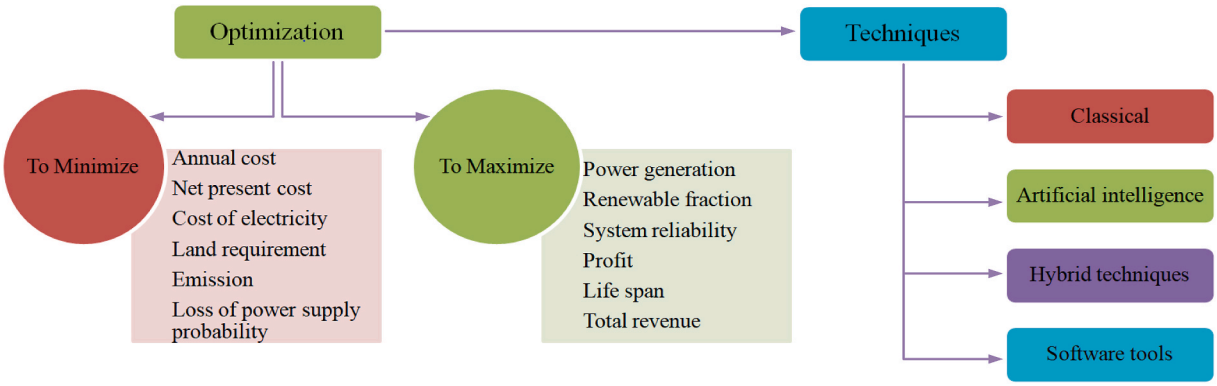


Fig. 3. Classification of optimization techniques.

system cost increases, and also, due to the system being installed in a remote area, the required operation and maintenance cost of all the components is high [60]. Linear programming technique has an important advantage which gives optimal results at a short interval, but, increasing the variables and dimensions of the function may not allow us to determine the optimal one. In analytical methods, additional computations are required to carry out optimization, and also, the most absolute solutions are not around to converge [61].

3.2. Graphical construction method applied to optimal sizing of HRES

Using graphic methods, it is easy to solve any optimization programming problem consisting of only two variables. The graphical construction method often more difficult to find a flexible solution and various approximations are necessary to explore. However, the overall procedure is easy to understand and does not involve any complexity [62]. This approach does not deal with high-dimensional problems with too many variables. Although contributes certain limits, both graphical and mathematical approaches may be prioritized for various practical applications if utilize properly [63]. A graphical technique is proposed to decompose CO_2 emissions in multi-product plants, as well as technological and economic feasibility [64]. A graphical user interface model using the time series method is utilized to optimize the off-grid hybrid system based on minimum life cycle cost and analyzed the tradeoff between energy storage and diesel fuel generator usage [65]. A geometrical construction method based on long time series of solar radiation is employed to analyze the sizing of the PV system, which would have been expected to deliver continuous power to load [66].

3.3. Iterative method applied to optimal sizing of HRES

A mathematical technique that produces approximate solutions to problems is known as an iterative process. An iterative algorithm generally employs the application of looping structures, which iterate over a sequence of needed steps to generate the desired output. It is a recursive method that stops when the right configuration according to requirements is found. Iterative approaches are used in renewable energy hybrid research from the beginning to model and optimize [62]. Iterative approaches are simple and easy to execute, but they consume a considerable amount of time because multiple loops would be needed to solve for the maximum and the minimum number of components, making the calculation more difficult [63]. An iterative optimization-based optimal sizing model for optimizing the capacity sizes of HRES (PV/wind) components that are installed at a location CDER (Bouzaréah) in Algeria to supply residential household appliances is recommended. The recommended model for the optimization process is comprised of three sub-models: sub-models of the hybrid system, a technical model based on DPSP and excess generated power to improve system reliability and an economic model based on indicators LUEC, TAC, and break-even distance. The hybrid system's best configurations are calculated in terms of the system's various desired stability criteria and the most cost-effective configuration [67]. A statistical model is utilized to calculate solar and wind power and, an Iterative procedure for optimal sizing of solar PV, wind, and the battery has been carried out and compared under different scenarios for different LPSP and a different number of consecutive hours [68].

3.4. Probabilistic method applied to optimal sizing of HRES

A probabilistic method is based on the theory of probability. It can employ implausibly, but using these effects to decide the optimum solution does not guarantee it. For sizing components of a system, consideration is usually taken into account for one or two system performance indicators. Probabilistic methods deal with multi-objective functions, non-linear system element response characteristics, as well as a long-time sequence of weather variables. An analytical approach is used to convolute the probability density function of the power generated by solar and wind-generating systems. A new probabilistic method was applied to the optimal sizing of hybrid solar-wind generation systems that allows for estimating the long-term average performance of a hybrid solar–wind power system [69]. A hybrid programming model integrating probabilistic analysis developed based on the MCS approach and ANN training embedded in a genetic algorithm optimization model, to ascertain the key behavior characteristics of PWHS effectively under unpredictable environmental circumstances [70]. A novel probabilistic approach is introduced to optimize an off-grid hybrid energy system (PV/WT) and to estimate the probability distribution of power generated from a designed system installed in the South China Sea. And also, evaluates the battery capacity of the hybrid system using the battery level coefficient mode. Sensitivity and optimization studies show that LPS 1% ought not to be used as a globally applicable measure of performance around the South China Sea [71]. The probabilistic approach based on a closed-form solution approach is developed to estimate the effect of the solar tracking system on HSWPS output performance. Reliability analysis has been performed using EIR index for varying annual load demand time patterns [72].

3.5. Analytical methods applied to optimal sizing of HRES

To estimate the component size as a function of its economic viability, analytical methods use computational models based on mathematical and theoretical analysis and calculations. It is a reliable method that requires only information from historical data. An analytical and IPSO are employed to calculate the optimal sizing of a standalone hybrid PV-wind-battery system to minimize the system's total annual cost [73]. A novel analytical method is employed to extract unknown parameters of the single-diode PV model directly based on the manufacturer's datasheet values using explicit equations [74]. An

Table 1
List of a few optimization techniques and their inspiration, merits, and demerits.

Algorithm	Inspiration	Advantages	Disadvantages	Ref.
Genetic algorithms (1960)	Population bases optimization Darwin's reproduction and survival of the fittest theory Process of natural selection	Can be applied with both discrete and continuous parameters Suitable for solving multi-objective problems Quite robust	Convergence speed is slow Knowledge of the problem is discarded once the population changes Can be trapped into local optima Can be inaccurate	[16]
Cross entropy (1997)	Adaptive importance sampling procedure	Desirable asymptotic properties Faster convergence Strong exploitation ability High robustness large probability of finding the global optimization	No guarantee that it will not get trapped in a local minimum	[17, 18]
Simulated annealing (1983)	Annealing as a metallurgic process Stochastic gradient search	Capable of escaping in small probability and worse solutions during its iterations Time-consuming Ability to avoid becoming trapped in local minima	Cannot guarantee to produce the exact optimum	[19, 20]
Particle swarm optimization (1995)	Population bases optimization Social behavior of organisms such as fish schooling and bird flocking	Fast convergence Knowledge of good solutions is retained by all the particles	Premature convergence as it lacks diversity	[16]
Artificial Bee Colony (2005)	Intelligent behavior of honey bee swarm	Ability to get out of a local minimum Efficiently used for multivariable and multimodal Superior to other derivative-free optimization	limited local search ability and Slow convergence speed over some constrained problems and complex functions	[21]
Biogeography-based optimization (2008)	Population-based evolutionary Migration of species between different habitats	Simplicity Efficiency Fewer computational steps per iteration	Provides an optimal local solution Performance degrades rapidly as the dimensionality of the search space increases	[22, 23]
Flower pollination (2012)	The natural process of flower pollination The physiological process of mating in plants	Strong searching ability Requires Few parameters Simple structure	Premature convergence Low precision Slow convergence speed Involuntary orientation toward a bad solution. Inadequacy in terms of intensification and diversification of new solutions	[24]
Social Spider Optimizer (2013)	Inspired by the social spider cooperative behavior strategy	Better optimal solutions Smaller standard deviation SSO possesses three new solution generations per iteration	Complicated computation Long time for tuning control parameters Low convergence speed	[25]
Grey wolf optimizer (2014)	Social leadership hierarchy and hunting behavior of grey wolves	Simplicity Requires fewer control parameters Ease of implementation	Suffers from the lack of the population diversity An imbalance between the exploitation and exploration Premature convergence	[26]
Jaya Algorithm (2016)	Survival of the fittest principle	Easy to implement Requires only common control parameters Execution does not require any algorithmic-specific parameters Conceptual and computational simplicity	The problem of premature convergence Trapped in local optimums	[27]
Dragonfly algorithm (2016)	The natural behavior of the artificial dragonflies	Few control parameters The information about the position of the individuals can be maintained at the time of iterations Memory space is less utilized Effectiveness and successfulness	Premature convergence in some complex optimization algorithms Control parameters tuning No theoretical convergence property	[28]
Pity Beetle Algorithm (2018)	Aggregation behavior, searching of Pityogenes chalcographus	Robustness and Efficiency Good scalability and adaptability Not affected by its algorithmic parameters The internal computational effort needed is minimal The increase in dimensionality has no noticeable effect on it.	Some flaws in terms of convergence speed and accuracy.	[29, 30]
Coyote Optimization (2018)	Population-based artificial intelligence Adaptation behavior of the coyote	Provides new mechanisms for balancing the exploration and exploitation in the optimization process	Premature convergence problem	[31]
Deer hunting optimization (2019)	Hunting behavior of humans toward deer	Attain the global optimum easily Can solve the problems excellent Highly balance the exploration and exploitation phases High searching ability	Need plenty of random parameters Sometimes attain Premature convergence	[32, 33]
Forensic-based investigation algorithm (2020)	Suspect investigation–location–pursuit process	Does not require the tuning parameters to be preset Faster convergence Shorter computational time Accelerates the process of determining the optimal solution	The upgrade is required to make the algorithm completely parameter-free, without population size and termination requirements. Recommended improving convergence time.	[34, 35]

(*continued on next page*)

Table 1 (*continued*)

Algorithm	Inspiration	Advantages	Disadvantages	Ref.
		Solves high-dimensional problems efficiently		
Manta ray foraging (2020)	Intelligent behaviors of manta rays	Effectively improve the optimization ability Regularly update its position Smoothly transit from exploratory to exploitative search Strong competitiveness in exploitation Easy to implement Require few parameters	Time complexity Not good at the fine-tuning of solutions around optima Slow convergence speed	[36, 37]
Golden eagle optimizer (2020)	The spiral trajectory of the golden eagle for hunting	Effectively determine global optimum and avoid local optima Solve problems with multiple nonlinear constraints efficiently Capable of exploring the landscape through intense Abrupt movements in the initial stages of the search good exploration Does not shrink during iterations of the main loop	GEO converges a little later than other algorithms in initial iterations	[38]
Tunicate Swarm (2020)	Bio-inspired artificial intelligence algorithm Swarm behavior of tunicate	Inspiration and performance are effectively proven over several problems Great performance for different constrained and unconstrained engineering design problems	High intensification in the search space Diversification of the search agents requires upholding	[39]
Jellyfish optimizer (2021)	The social movement of jellyfish	Require fewest evaluations Superior optimization algorithm in exploring the search space Excellent exploitation capability Requires less time and faster convergence	Slow convergence Tendency to become trapped at local optima	[40]

Table 2
Economic, reliability, social, and environmental indices, and their main formulas.

Assessment	Indicator	Description	Equation	Ref.
Economic	Levelized cost of energy	Required cost to produce one unit of energy	$\mathrm{LCOE} = \frac{\text{Total life cycle cost of the system}}{\text{Total annual energy generation of the system}}$	[41]
	Penalty cost for carbon pollutants	It calculates the penalty cost of CO_2 emission by DG	$PC = CO2W \,.\, CO$	[42]
	Annual Levelized Cost	Investigate the system's economic viability	$\mathrm{ALC} = \text{Annual capital cost} + \text{Annual O\&M cost} + \text{Annual Replacement cost} + \mathit{Annual\ Fuel} \cos t$	[43]
Reliability	Deficit power probability	The ratio of the power deficiency to the load power of the system	$DPP = \frac{\sum_{t=1}^{T}[DP\ (t)]}{\sum_{t=1}^{T}[Pload\ (t)]}$	[44]
	Hourly interruption probability	The ratio of the interruption of power at a time to the total annual hours of the load	$HIP = \frac{\sum_{t=1}^{T}[Time(Pavailable(t) < PLoad(t)]}{T}$	[44]
	Loss of load expected	The average number of hours for which the system load is expected to exceed the available generation capacity	$\mathrm{LOLE} = \sum_{i=1}^{8760}\mathrm{toutage(i)}$	[45]
	Expected energy not supplied	The energy that is not going to be supplied when the system load exceeds the available generation capacity	$\mathrm{EENS} = \sum_{i=1}^{8760}\mathrm{Eunserved(i)}$	[45]
	Load Dissatisfaction Rate	The ratio of the energy deficiency at an hour (t) to the required energy of the system	$\mathrm{LDR} = \frac{\sum_{t=i}^{T}(Eload\ (t) - \mathrm{Esupply\ (t)})}{\sum_{t=1}^{T}Eload\ (t)}\%$	[46]
	Energy index of reliability	Measure the reliability of each candidate hybrid system design	$EIR = 1 - \frac{\mathrm{EENS}}{\mathrm{E}}$	[47]
	Loss of power supply probability	Measures the probability of insufficient operation of the hybrid system when it fails to satisfy the energy demand.	$LPSP = \frac{\sum_{t=1}^{8760}LPS(t)}{\sum_{t=1}^{8760}Pload(t) \times \Delta t}$	[48]
	Equivalent loss factor	The ratio between the loss of load and the demanded load	$ELF = \frac{1}{T} \times \sum_{t=1}^{T}\frac{Q(t)}{P(t)}$	[49]
	Excess energy fraction	The ratio of the wasted power at a time to the total generated power of the system	$EEF = \frac{\sum_{t=1}^{T}Wasted\ energy\ (\mathrm{t})\ .\ \Delta\mathrm{t}}{\sum_{t=1}^{T}Generated\ energy(t)\ .\ \Delta\mathrm{t}}$	[50]
	Level of autonomy	Load acted as a percentage during the system's operating time	$\mathrm{LA} = 1 - \frac{\text{The number of hours of load failure}}{\text{Total hours of operation}}$	[51]
	Novel grid power absorption probability	The probability that the system needs to purchase electricity from the utility grid when the renewable energy is unable to feed the load	$GPAP = \frac{\sum_{t=1}^{T}Power\ purchased\ from\ grid}{\sum_{t=1}^{T}\text{load demand}}$	[52]
Social	Human development index	To scale a nation's general achievement in its social and financial measures	$HDI = 0.091\ln[\mathrm{P}_{LY/C}] + 0.0724\ \mathrm{HDIy,z}$	[42]
	Job creation	Use of native resources leading to job increases	$JCy,z = \text{JC Renewable sources} \,.\, \text{PRenewable sources} + \text{JCDG} \,.\, \text{EDG} + JCBattery \,.\, EBattery$	[42]
Environment	Renewable fraction	Electrical energy supplied to the required load demand through the renewable energy sources	$\%\ RF = \left(1 - \frac{\sum \text{P Diesel generator}}{\sum \text{P Renewable sources}}\right) \times 100$	[42]
	Emissions of CO_2	The total amount of CO_2 generated by Diesel generators	$Et = \sum_{t\epsilon T}\sum_{n\epsilon N}Engnnt$	[53]

efficient analytical approach is presented to model a small autonomous power system (DG-wind–solar) which accounts for uncertainties and outages of sources, and obtained results are compared with Monte Carlo simulation in terms of accuracy and computational time [75]. The availability of the excess amount of energy from the solar system based on the concept of solar radiation usability is determined. An analytical model is emphasized to predict how the performance of renewable energy units based on hydrogen storage strongly depends on the parameters of the solar array and turbine generators, meteorological data, and load profile [76]. In addition to the above methods, there are several other approaches are described in various publications to design and optimize hybrid renewable system which includes the trade-off method introduced for solving multi-objective under uncertainty which intended to design standalone renewable power sources [77], Scenario analysis technique is applied to size optimal off-grid Solar PV system [78], Knowledge-based approach to minimize total capital cost is detailed out in Refs. [79,80]. The merits and demerits of classical methods are presented in Table 3. The summary of the classical methods is presented in Table 4.

During the year 1990–2000, several papers applied only classical techniques to determine hybrid renewable system sizing. Every time interval, typically an hourly or daily period, is simulated using numerical approaches, and due to the linear changes in the decision variables, suboptimal decisions will arise. The simplicity of the size computations makes the analytical approach advantageous, but it has the drawback of making it difficult to estimate the location-dependent mathematical equations' coefficients. The graphical construction technique is an easily understandable one with no complexity, but due to the requirement of various approximations, it is not flexible. The main strength of the classical techniques are easy to understand and apply, but suffers from several disadvantages, such as the requirement of much more computational effort, rigid iterations, weak flexibility, poor convergence, more computation time, and the inability to measure dynamically shifting measures. Researchers have recently used artificial intelligence to handle hybrid renewable energy system optimization challenges. These techniques are effective for this problem because it can reach the global optimum.

4. Artificial intelligence based optimization technique applied to optimal sizing of HRES

Artificial intelligence is a discipline of computer science that learns and creates intelligent machines and software. The phrase "artificial intelligence" refers to a machine or artifact to carries out similar types of tasks to those that characterize human cognition. It is defined as "the study and creation of intelligent agents," in which an intelligent agent pursues measures that raise its chances of success [81]. Artificial intelligence algorithm has been defined as a set of instructions to be followed in operations. The proper use of intelligent techniques results in useful systems with increased performance or other features that traditional approaches cannot provide. Optimization is the problem of finding a set of inputs to an objective function under constraints those results in a maximum or minimum function evaluation. Under specific restrictions and acceptances, the optimum solution to the problem is selected from among the optimization candidates, and one can also be able to perceive how humanistic factors are connected to behaviors while considering the procedures involved in optimization. Because of this, artificial intelligence has shown itself to be the ultimate option for optimization issues. It is sufficient to comprehend the point at which optimization becomes the subject of decision-making, Artificial intelligence, and optimizations. It is also feasible to discuss and evaluate the expressed relationship from a more technical standpoint. Artificial intelligence is based on a collection of broad techniques and methods for transferring intended logical and mathematical processes to reality realistically and flexibly. The primary reasons why the Artificial intelligence field is considered for optimization problems are because classical optimization techniques are not sufficient for advanced optimization problems, the Artificial intelligence techniques is considered for optimization problems due to its effectiveness, speed, flexibility, feasible solutions, and efficiency [82]. Artificial intelligence-based optimization algorithms are population-based and are generally classified based on swarm intelligence, physics, humans, and evolution, as shown in Fig. 4 below. Algorithms inspired by the social behavior of animals are grouped into swarm intelligence (PSO, ABC, FA, etc.); algorithms inspired by the physics law are grouped into "physics-based" algorithms (SA, GSA, etc.);

Table 3
Merits and demerits of classical methods.

Types	Description	Merits	Demerits	Ref.
Analytical	It is focused on an interpretation of historical load demand curves or statistical data. No dedicated mathematical optimization tools were employed. Computational models are used to classify the components of interconnected energy systems to determine the system's feasibility.	Very simple to calculate sizing In comparison to MCS, it is more computationally efficient and consumes less computing time.	Estimating the coefficients of location-dependent mathematical equations is difficult Need longer computation time Less flexibility in systems design For each section, it necessitates the use of discrete-continuous probability density functions	[74, 87]
Numerical	Simulations are performed at regular time intervals, which are normally on an hourly or a daily basis	There is no specified procedure for obtaining a series of approximated solutions.	Since the computation involves linear shifts in the decision variables, suboptimal solutions are found Need longer computation time	[74, 88]
Probabilistic	Based on random variables depends on the performance of a system.	Simplest sizing methodologies Eliminate the need for time-series data	These techniques produce results that are not optimal for determining the optimal solution Only one or two system performance indicators are considered for optimization The dynamic changes in the hybrid configuration cannot be reflected.	[74, 89] [90, 91]
Graphical Construction	It displays a graphical solution to an optimization constraint.	Easy to understand	Generally, graphical approaches optimize only two parameters (photovoltaic and battery, or photovoltaic and wind turbine). Certain critical parameters are entirely ignored, such as the PV module slope angle and wind turbine assembly height.	[51, 89]
Iterative	A recursive program is used to evaluate the performance of interconnected energy systems, which ends when the optimal system design is reached	Easy to code Capability to trace the threats at early phases	This method does not optimize the area and slope angle of PV modules, wind turbine swept area or installation heights. Increased computational efforts and suboptimal solutions	[51]

Table 4
Summary of studies based on classical methods [92–100].

Method	Energy sources	SA/GC	Location	Optimal indicators	Objective function(s)	Findings	Ref.
Stochastic	PV/Battery system	SA	Brazil	Reliability Economic	Minimize LPSP Minimum TLCC	In comparison to deterministic approaches, stochastic methods are more realistic, economically feasible, and more accurately predict system operation	[92]
Graphical	PV/WT/Battery	SA	Massachusetts	Economic	Minimum the cost	For a given load demand and a level of reliability, the optimum size of batteries and PV modules were calculated based on the economic approach of the power system.	[93]
Numerical	PV system	SA	Turkey	Economic	Minimize LCC	For a certain LLP level, the life cycle cost differs significantly The proposed system's energy and CO_2 payback periods are 6.2 and 4.6 years, respectively	[94]
Numerical iterative	PV/WT/BM/BG/MHP	SA	India	Reliability Economic Energy conversion	Minimize COE	The proposed IREOM model is entirely reliant on renewable energy sources, reduces the need for traditional energy sources, and balances energy needs with a sufficient energy supply For 0.999 EIR: COE ($/kWh) 6.80; For 0.99 EIR: COE ($/kWh) 6.40	[95]
Iterative	PV/WT/Battery	SA	Ege University	Economic	Minimize LCOE	Sizing variables and performance variables are identified according to the objective function under constraints. PV/WT/BS: COE ($/kWh) 0.89, CO_2 emission (Equiv./kWh) 82 Grid: COE ($/kWh) 0.23, CO_2 emission (Equiv./kWh) 493	[96]
Probabilistic	PV/WT/Battery	SA	Dharwad	Reliability	Maximize reliability	The utmost utilization of energy sources is analyzed by evaluating different reliability indices The results of the spring season indicate maximum reliability for all reliability indexes, followed by summer, winter, and finally autumn.	[97]
Probabilistic	WT/Battery	SA	Kythnos-Greece	Reliability	Maximize reliability	The Loss of power supply probability and expected unserved energy are calculated to assess the planned system's reliability Chronological: LOLP (Hrs/yr) 2328, EUE (kWh/yr) 37055 Probabilistic: LOLP (Hrs/yr) 2475, EUE (kWh/yr) 33707	[98]
Analytical	PV/WT/Battery/UPS	SA	Corsica Island	Technical Economic	Minimize LCOE	The LCE for hybrid energy system configurations that achieve the optimal LPSP is highly dependent on the efficiency of available renewable energy.	[99]
Analytic	PV system	SA	Spanish	Reliability Technical	Minimize LLP Capacity of PV	The proposed new model is accurate as a numerical and simple as an analytic model.	[100]

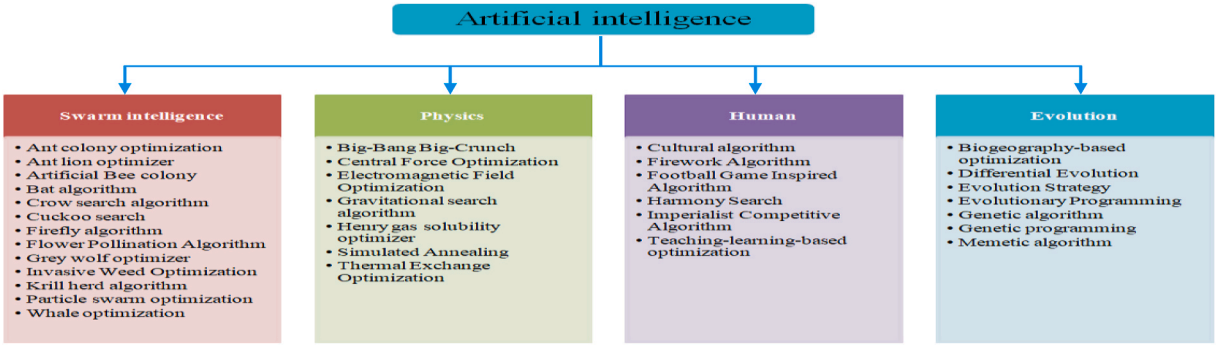


Fig. 4. Classification of Artificial intelligence techniques.

algorithms inspired by the social behavior of humans are grouped into "human-based" algorithms (TLBO, HS, TS, etc.); and finally, evolution-based algorithms (GA, DE, etc.) are based on the principle of Darwin's evolutionary theory [83].

The increasing complexity and challenges in real-world problems have necessitated the development of more efficient optimization methods, notably artificial intelligence optimization algorithms, in recent decades. Numerous random search algorithms and artificial intelligence based optimization algorithms have been proposed for optimization in recent years. Artificial intelligence based algorithms solve problems of optimization stochastically with some prior knowledge about random search. AI is widely used because it is simple to comprehend and execute, and it performs well in practice. Artificial intelligence based optimization algorithms are widely employed due to their ability to handle non-smooth and nonlinear problems, and faster convergence. An important case that must be taken into account in artificial intelligence is the dynamic balance between two conflicting criteria such as diversification and intensification. Diversification entails developing diverse solutions to fully explore the global search space, while Intensification refers to the process of focusing the search on a local area based on exploiting the information that a solution is currently being found in this search space [84]. Additionally, it is worth noting that every artificial intelligence optimization algorithm can have several advantages and disadvantages; hence, searching for the optimal algorithm can be deemed futile. Any algorithm's effectiveness is determined by the nature of the problem it is used to solve. Thus, the optimization problems are generally resolved with several algorithms to identify the algorithm that yields the best effect. Solving the issues related to the

design of energy hybrid systems necessitates a robust approach with high optimization speed and precision, to assess the optimum capability of system components with the lowest cost and highest efficiency. Various algorithms have been developed and implemented recently to determine the feasibility and optimum configuration of HRESs to energize a variety of load categories in a variety of locations in the country. The summary of the single algorithm used to solve size optimization of HRES is presented in Table 5. Optimal sizing and energy capacity estimation of a stand-alone PV array-battery bank-hydrogen system by using a genetic algorithm for different locations with different climatic conditions are determined. The goal of this work is to reduce overall costs while restricting the probability of loss of energy supply and the state of charge for the long-term storage aspect. The research on the impacts of the local atmosphere has included the use of local meteorological data concerning determining the thermal requirements and solar power generation. The effects of the sizing comparisons between locations show that the local environment has a significant effect on the final price, by significantly altering the PV resource and the building's thermal requirements [85]. A new bio-inspired optimization technique for optimal locating and sizing of the BESS in 48 bus distribution systems is proposed. The analysis was performed on four different scenarios based on the number of energy storages and the presence/absence of solar PV. Based on an improved cayote optimization, the optimization was performed on a 48-bus distributed system to minimize overall system losses, and the results showed that the proposed algorithm is more efficient than FA,WOA, PSO [86].

An integrated energy system is designed to meet the cooling demand for a hospital in India, consisting of a solar thermal evacuated plate collector, biomass-fueled heater, and heat and storage tank. This study modeled and compares various configurations, including three types of biomass-based heaters with varying efficiency and costs, as well as three types of local biomass. The genetic algorithm was used to minimize the levelized expense of cooling and the carbon dioxide pollution index simultaneously for the optimal case [101]. A 3 E fuzzy decision-making optimization technique is adopted for the design and evaluation of a hybrid solar/wind/diesel generator energy system, with the options of battery storage and pumped hydro storage technologies. The primary objective of this work is to supply the required electricity with the lowest possible cost of electricity to a large-scale RO project, which is to supply the regular freshwater requirements of the Pharma peach resort in Baltim, Egypt [102].

Recently, several effective strategies are being employed to treat wastewater from municipal and industrial discharges to attain safe aquatic ecosystems. A membrane bioreactor seems to be an alternative way to reuse water in wastewater treatment. A nitrifying-enriched activated sludge (NAS) was approached to improve the performance of the membrane bioreactor by controlling the microbial population, which lowers fouling [103]. By monitoring the rate of nitrate accumulation, the ecophysiological interaction between the ammonia-oxidizing and nitrite-oxidizing bacteria communities has been investigated to reduce metabolite levels and mitigate CO_2 [104]. Also, the autotrophic interaction has been dealt with between Chlorella Vulgaris and NAS to investigate nutrient removal, carbon capture, and metabolite generation in the treatment of wastewater plants [105]. Environmental pollution arising in the water sector has become a contemporary problem for sustainable development. Membrane bioreactor technology is widely used for municipal and industrial wastewater treatment and has several advantages over conventional treatment plants. Membrane bioreactors have not been widely used because fouling requires regular cleaning or replacement which comes at a high economic cost, and also due to physicochemical interactions energy consumption will be more. As a result, integrating renewable energy sources and membrane bioreactor systems will reduce costs and operational issues while also significantly increasing MBR application. An optimal grid-connected hybrid renewable energy system (PV-WT-Battery) has been proposed using a novel dual-scale optimization approach to meet the dynamic load demand of membrane bioreactor systems. The optimal system component configuration, combined environmental cost (CEEC), and CO_2 emission are determined, and when the grid supplies 53% of the membrane bioreactor load and a designed hybrid renewable energy system supplies the remaining 47%, the lowest CEEC is obtained [106]. For the stable operation of a decentralized wastewater treatment plant that includes a plug-flow step-feed bioreactor, a hybrid power system consisting of solar and wind energy was designed, and under various climatic conditions, the efficiency of power generation and system generation were analyzed [107]. The summary of comparison among various optimization algorithms applied to optimize HRES is illustrated in Table 6.

It has been observed that the majority of hybridized RE sources are typically PV, wind, and DG paired with energy storage to increase stability and reliability. The production of renewable energy (PV and wind) is highly dependent on weather conditions, which are not predictable with the precision necessary for electricity supply. To circumvent this limitation, PV and wind-based renewable sources can be paired with a more predictable energy source, such as biomass fuel, which can be delivered on demand. The key factor for using biomass fuel is that it is most plentiful and cheaply accessible in the form of animal dung, and farm waste. On days with rain or days without wind, the overall electricity generation would always be dependent on biomass. Numerous researchers have been examining the viability of biomass-based energy systems for rural electrification in recent years. Biogas-solar energy hybrids reduce complexity, maintain the lowest unit cost, have a quick response, and allow for easy optimization and regulation. However, for effective hybrid biogas-solar technology, an optimization model must be developed to help determine the location for the implementation of the system. Different hybrid wood chip CHP system cases with solar photovoltaic, battery and diesel systems are designed to assist villages in improving their living standards by guaranteeing their electricity and thermal needs and reducing wood consumption to save forests and the national wood deficit. The results show that grid extension is not a good choice for powering rural areas, whereas the hybrid design CHP/PV/Diesel generator is the best choice in terms of both economics and the environment [108]. The optimal evaluation of battery energy technology is one of the key solutions to mitigate the intermittent and variable nature of renewable generation. The authors investigated the impact of different storage technologies on optimal asset configuration to reduce generation costs and carbon footprint emissions. Techno-economic analyses have been performed using mixed-integer linear programming for hybrid solar/biomass/DG/battery for the application of six different case studies in India. Results show the initial cost of Li-ion technology is high, but it has a lower LCOE, is more efficient, and requires less maintenance than valve-regulated lead-acid technology. Flooded lead acid is also cost-effective, but losses are high, efficiency is low, and it requires high maintenance [109]. Thermal energy storage is incorporated with the system, to save thermal energy that will be used during hours without solar radiation. Performance analysis of an innovatively modeled tri-source RE-powered comprehensive system consisting of solar, wind, and biomass was performed to achieve good energy and exergy performance. The overall energy and exergy were presented as 64.91%–71.06% and 31.80%–53.81%, respectively, for four different scenarios involving multiple useful products [110]. A comprehensive term economic analysis on biomass-geothermal integrated systems using the genetic algorithm to utilize the desalination plant has been performed. An integrated geothermal system is proposed for the application of freshwater production from seawater and electricity generation in Turkey, resourced by harvesting geothermal energy and heat from the combustion of municipal solid waste. Three objective functions such as energy efficiency, product cost rate, and CO_2 emission rate were considered simultaneously, and the emission rate has been considered to be important to optimize [111]. A biomass-assisted multi-generation integrated plant has designed and evaluated the techno-economic assessment of the system. The multigeneration process includes power, hydrogen, heating, cooling, clean water, and drying; also, the generated

Table 5
Summary of studies that used a single artificial intelligence techniques to optimize HRES.

Algorithm	System components	SA/ GC	Analysis	Location/ Application	Objective function	Constraints/ Variables	Findings	RRef.
DMGWO	PV/WT/BT	SA	Economic Reliability Energy	Jaisalmer, India/ Telecom Towers	To optimize the size of HRES by minimizing LPSP, LCOE, EEG	Area of PV panels Wind and battery capacity, Number of AD Solar irradiance and wind speed Telecom base station load Normalized traffic at the given time Power balance	The designed system for telecom applications is dependable, with an LCOE of $0.12, which is less than the $0.33 of diesel-powered systems The impacts of decision variables are highly correlated among the objectives	[115]
MOCSA	PV/DG/FC/ HT/EL	SA	Techno-economic Environment	Kerman, Iran/ To electrify community load	Minimization of TNPC and LPSP	Number of HTs, ELs FCs, DGs and PV panels	For 0% and 10% of LPSP, the total system net present cost of PV/DG with fuel storage is less than the cost of PV/DG system, concerning without considering the impacts of the operating reserve, uncertainty, and emission	[116]
Fuzzy-TOPSIS	PV/WT/BT/ HS	SA	Economic Reliability Environmental	Vietnam, South Asia/ The wastewater treatment plant	To optimize the size of the components To compute feasible electrical storage capacity To calculate TAC, COE, ACC,AOMC, and ARC	Loss of load probability Loss of loading energy Loss of energy supply	The optimum configuration of HRES was 165 photovoltaic solar panels and five wind turbines, which consumed 90% and 10% of DC and AC power, respectively, and was proposed as the best option for meeting reliability, economic, and environmental requirements The reliability of the systems to meeting dynamic demands is high, with up to 86% of electric loading	[117]
GA	PV/WT/BT/ DG	SA	Economic Technical	Arizona, USA \ Reverse osmosis desalination	To minimize LCOE	Loss of power supply probability Types of wind turbine generator Number of wind turbine generators PVs tilt angle Number of PVs string in parallel Number of battery strings in parallel Number of Diesel generators	The optimal system for the run time of DG in an hour every day consists of PV/WT/DG/battery with the lowest COE at 0.56 USD/kWh and the LCOE has been found at 3.84 USD/m^3 For DG run time between 9 a.m. and 9 p.m. every day, a wind/photovoltaic/battery hybrid system with LCOE of 0.69 USD/kWh and LCOW of 4.48 USD/m^3 has been found as the optimal design	[118]
Jaya	PV/WT/BT	SA	Economic Technical	Rafsanjan, Iran \ Consumer load demand	To minimize TAC	Maximum allowable LPSP Maximum number of PVs and WTs	For various LPSP values, the results indicate that the PV/WT/Battery system is the most effective and economical option, followed by the PV-Battery and WT-Battery systems	[119]
TSA	PV/WT/BT/ DG	SA	Techno-economic	Abu-Monqar, Egypt/ Small village	To minimize energy production cost To maximize system reliability	Number of installed PV panels Number of installed wind turbines Number of diesel generators Number of installed Batteries Battery SOC	The PV/WT/SB/DG system will provide electricity to the small village with an LCOE of $0.339,264/kWh, and a NPC of $9935235	[120]
MILP	PV/WT/BT	SA	Technical Economical	South-Tyrol, Italy/ Mountain Lodge	To minimize the overall cost of the system	Number of installed PV panels Number of installed wind turbines Number of installed Batteries Amount of energy delivered or absorbed by batteries	The optimal solution to the problem consists of 86 numbers of PV panels with an installed capacity of 20.64 kWp, 1 unit of wind turbine with 10 KW and 30 kWh battery capacities to minimize the objective function	[60]

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Table 5 (*continued*)

Algorithm	System components	SA/ GC	Analysis	Location/ Application	Objective function	Constraints/ Variables	Findings	RRef.
Improved numerical algorithm	PV/BT/GT	SA	Economic	Perak, Malaysia/ Universitiy Teknologi	To minimize annual total load cycle cost	Loss of load probability	The proposed PV/GTG solution will save USD 2.1 million in annual operating costs when compared to the current gas-turbine-based generator, while the power grid link model will save USD 1.16 million in annual costs	[121]
PIO	PV/WT/BT	GC	Economic Technical	Boston, USA/ Logan International Airport	To minimize LCOE	Loss of power supply probability Power Fluctuations Battery SOC Charging and discharging of battery Operating reserve	Demand side management has a beneficial impact on device size and means that the amount of electrical storage elements required has been greatly reduced, leading to lower cost per electricity unit	[122]
NSGA-II	PV/WT/BT	SA	Economic	Tunisia/ Remote area	To minimize the cost of the system, LPSP	Number of PV arrays Number of WT Number of batteries State of charge of batteries Loss of Power Supply Probability	These results show that the design comprised of a hybrid PV/WT and electrochemical storage (5 units of WT,28 units of PVs, and 9 units of batteries) with low depth of discharge and low cyclability, is more economical and the most relevant configuration in terms of the system cost and LPSP	[123]
MDPSO	PV/WT/BT/ DG	SA	Economic Reliability	Essaouira, Morocco/ Small fishing village	Reliable power supply at minimum LCOE Increase the reliability based on LPSP Reduce CO2 emissions, Maximize renewable fraction	Number of PVs Number of WTs Number of DGs, Number of Autonomy days	Renewable energy can be considered as a good alternative to enhancing energy access in selected areas as the solution is very promising in terms of achieving values of the LCOE (0.17 $/kWh), LPSP (0.12%), and REF (90%)	[124]
MOSaDE	PV/WT/BT/ DG	SA	Economic Reliability	Yanbu, Saudi Arabia/ For household applications	To analyze COE, LPSP	Renewable Factor Nominal power of PV Number of autonomy days Number of wind turbines	The results of this study are useful in investigating the optimal scheduling of HMS components and can be used as a powerful reference for the economic operation of PV and wind turbine generators.	[125]
IACA	PV/WT/BT/ HS	SA	Economic Reliability	Zhejiang, China/ Remote island	Minimize annual system cost Maximize the reliability	Number of PV panels, Number of wind turbines, Number of batteries Number of hydrogen tanks Power balance, SOC of battery	From the simulation results for different LPSPs, it is revealed that decreasing the system cost and increasing the system reliability is paradoxical.	[126]
GA	PV/WT/BT/ DG	SA	Techno-economic Emission	Fanisau, Northern Nigeria/ Remote village	To minimize total annual cost	Number of PV, WT, Batteries Rated Capacity of diesel generator loss of power supply probability Renewable energy fraction	The COE was found at 0.25 USD/kWh, which is two and a half times the current electricity tariff in Nigeria.	[127]
GA	PV/WT/BT	SA	Economic	Near to Colorado, USA/ Residential remote area	Minimize system total cost	Area of PVs and WTs Battery capacity State of Battery charging and Discharging	The combination of PV and wind in a hybrid energy system with optimal sizes such as PV area (40.0004 m^2), wind turbine area (50 m^2), and battery capacity (30 KWh) is the most economical and reliable solution for electrifying remote area loads and reduces the battery bank and diesel requirements, lowering the system total cost.	[128]
MLUCA	PV/WT/BT/ DG	SA	Economic Environmental	Residential home/ Pacific Northwest	Minimize Total annual cost Minimize GHG emissions	Battery capacity Energy supply Component size	As the results are compared, the optimal PV/wind/diesel HPG system outperforms the renewable PV/wind HPG system	[129]

(*continued on next page*)

Table 5 (*continued*)

Algorithm	System components	SA/ GC	Analysis	Location/ Application	Objective function	Constraints/ Variables	Findings	RRef.
ABC	PV/WT/SB	GC	Techno-economic	Italy/ Residential load	To minimize net present value	System components Charging and discharging rate of battery	By using real-time weather and electrical data, an optimal microgrid has been designed and optimized, with a net present value of €38900.	[130]
MOPSO	PV/WT/BT	GC	Reliability Cost Environmental	Ismailia Governorate, Egypt/ Rural community	To minimize LPSP, COE To maximize RF	Number of PVs, WTs, and batteries Energy balance Battery storage limits	The scenario (selling and purchasing electricity from the grid) is the best as the total present cost of the system was $1,808,583 From an environmental aspect, it is clear that (selling electricity to the grid.) scenario is the best, with 381 tons of CO_2 emissions per year.	[131]
ACO	PV/WT/BT/ DG	SA	Technical Economical	Madhya Pradesh, India/ Remote area	To minimize TAC	Loss of power supply probability Number of wind turbines Number of PV panels Number of battery	The levelised cost of energy for an optimized sizing configuration of 23 units of 5 kW WTG, 12 units of 5 kW solar PV panel, 2 units of 5 kW DG and 150 units of 12 V, 100 Ah battery is Rs. 10.177 which is much less than the per unit cost of energy for DG	[132]
MOEA	PV/WT/BT/ DG	SA	Economic social	Tindouf/ Small community	To minimize NPC To maximize HDI and JC	Number of PV's, WT, and batteries The output power of DG, State of charge of battery	The optimal system, consisting of 2 kWp PV, 14.7 KW WT, 11KVA DG, and 75.84 Kwh Battery, is the most cost-effective solution, with an NPC of $125,465 and an LCE of 0.21 $/kWh.	[133]
ICA	PV/WT/FC/ HT/EL	SA/ GC	Economic Reliability Environmental	Northwest region of Iran/IEEE load profile	To minimize the total cost of the system To formulate the amount of CO_2 emission To address the cost of CO_2 emission	Loss of load expectation Loss of energy expectation Loss of power supply probability Equivalent loss factor Installation angle of PV array Energy stored in the HST	The optimally configured non-autonomous HGPS costs 92.6% less than the autonomous HGPS for a maximum purchased capacity of 50 kW, based on current prices, in exchange for 5778 tonnes of CO_2 emissions.	[134]
GA	PV/WT	GC	Economic Sensitive	Catalonia/ Rural township	To determine the minimum NPV	Area of PV modules Number of PV and WT modules PV technology capital cost Wind technology capital cost Electricity price Inflation rate PV module and wind turbine reference efficiencies interest rate	The proposed model responds well to adjustments in the input parameters and variables while presenting reliable sizing solutions. According to the findings, grid-connected photovoltaic (PV) and wind power technologies will be more cost-effective than not having HRES.	[135]
NSGA II	PV/WT/BT	SA	Economic Reliability	Kent, UK/ Household	To minimize the system's total cost To maximize reliability	Wind turbine rotor swept area PV panel area Number of batteries	By comparing the design methods, it can see that there are more hours in the year with very little chance of having power shortages in NSGA-II and chance-constrained programming than in NSGA-II and Monte Carlo simulation. It is also observed that the maximum hourly blackout occurrence probability is 0.6 in NSGA-II and MonteCarlo simulation, which is less than NSGA-II and around 0.8 for the MonteCarlo simulation.	[136]
PSO	PV/WT/BT/ DG	SA	Techno-economic	Morocco, Africa and Baghdad, Iraq/Commercial building	To determine the minimum NPC	Loss of power supply probability Renewable fraction	The resulting total showed that the cost of the project reached 31 K dollars and LCOE of 0.2330 $/kWh for the city of Baghdad, while the NPC touched 43 K dollars and LCOE of 0.3214 $/kWh for the city of Rabat	[137]

(*continued on next page*)

Table 5 (*continued*)

Algorithm	System components	SA/GC	Analysis	Location/Application	Objective function	Constraints/Variables	Findings	RRef.
PSOMCS	PV/WT/BT	SA	Technical Economic	Rafsanjan, Iran/Remote area	To minimize total annual cost	Uncertainty of load and resources Number of PV, WT, and batteries	From the simulation results, it is found that the WT/battery is the most cost-effective system in terms of minimum, maximum, and mean index for different months (January, April, July, and October) and the whole year	[138]

hydrogen in the gas phase has been stored under pressure. With the electricity and hydrogen generation rates increasing, the purchase rate of the proposed system increases. Furthermore, as the biomass input flow rate increases, so do the purchase costs of the biomass gasification subsystem. The energy and exergy efficiencies of the multigeneration plant were stated to be 63.84% and 59.2%, respectively [112]. Economic analysis has been proposed for microalgae biodiesel energy production, using highly energy-intensive technologies [113]. To make wind power more sustainable and reliable, the authors created a hybrid system that included a power generation section as well as a water supply section. A novel quintuple solar/-wind/hydro/biomass/geothermal renewable hybrid system along with a fuel cell has been designed to supply water and power. The wind turbine capacity factor is calculated to be 82.3%, with a unit electricity price of 0.161 $/kWh [114].

The authors compared six different layouts for producing substitute natural gas and electrolytic hydrogen from biomass using life cycle analysis. In all cases, the impacts of GEP, SED, and AP indicators views were carried out for this analysis. The results show that substitute natural gas derived from biomass and electricity has lower environmental burdens in terms of global warming potential and cumulative energy demand than conventional natural gas, despite having a higher acidification potential [173]. Techno-economic and energy analyses were carried out for off-grid hybrid PV, HKT, and BMG and batteries to supply electricity demand for a university located in Cuenca. For the biomass gasification process, wood, banana, sugarcane, and cocoa residues have been proposed, and this system has been simulated using Homer Pro under three control strategies [174]. A generic PV, BM, DG/battery model has been proposed and solved using mixed integer linear programming for four cases of load facilities. To mitigate the inherent nature of renewable power, energy storage systems play a key role, and various smoothing techniques were employed to provide smooth power output [175]. The ENBIOTEM model has been developed to investigate the techno-economic feasibility of agricultural residue biomass. The proposed model calculates the energy cost to be 71.6 $/MWh for a 300 MW optimal plant configuration. Numerous critical research issues remain unresolved in current hybrid solar-wind-biomass plant model optimization schemes [176]. The efficiency loss of biomass boilers at partial loads is one of the issues that have not been considered in a significant number of previous studies. Based on the installation and operation cost structure in existing literature regarding biomass-based hybrid systems, controversial cost structure issues have been identified, and cost learning curves are also not taken into account. While operating a biomass plant, a major obstacle is securing a continuous biomass supply with high quality and quantity. Efficient biomass supply chains and sufficient storage facilities are therefore needed to ensure timely biomass availability. Bio-fuel supply chains are linked to a variety of issues, including the economy, the environment, and society, as well as uncertainty and risks. To mitigate adverse ecosystem impacts and increase overall economic viability and social gains, robust modeling and optimization systems are needed to analyze and find sustainable alternatives for the design and function of biofuel supply chains simultaneously. In terms of the social component, job creation is the most significant aspect of the design and function of biofuel supply chains.

Power generation from water resources is highly efficient, inexpensive, and has fewer environmental effects than conventional energy. Currently, studies on hydro-related multi-energy power have arisen because of increased awareness of environmental protection and potential contamination caused by thermal generation. However, power generated from the hydro station has been affected by a range of complex constraints, including natural rainfall conditions, high construction costs, and difficulty increasing its capacity. To provide stable and reliable energy supplies, the inherent uncertainties associated with solar and hydro are a key challenge. By considering the seasonal and daily uncertainties of these energy resources, typical generation scheduling for nine scenarios has been proposed, to attain a reliable power supply and equilibrium power generation as an objective function. A short-term multi-objective optimization-based hybrid hydro-PV generation system is proposed, considering natural uncertainties and policy impacts that successfully maximize power supply reliability and electricity sales revenue simultaneously. Multi-objective optimization has been reduced to single-objective optimization using the $\in$-constraint method, and the LINGO solver is used to find the best solution. The following propositions are observations based on the calculated results: an equilibrium strategy increases the power supply economy and reliability, improves flexibility, each resource compensates with each other, and the structure of the energy supply can be adjusted based on subsidy policies [177]. Non-dominated Sorting Genetic Algorithms-II is used to optimize a multi-objective model for a hybrid hydro-PV power system with simultaneously considering energy generation and consumption, to minimize energy waste. The case study was conducted in Qinghai Province, where the world's largest hydro-photovoltaic hybrid system is available. With the consideration of reservoir release as a controllable variable, the maximum energy generation is attained. The results suggest that during dry and normal years, more attention needs to be paid to reservoir regulation [178]. The integration of dispatchable and non-dispatchable power is very crucial to effectively utilizing the resources. A coordinated optimization strategy has been proposed for a large-scale hybrid PV system and addressed based on the GFM-MOEA to maximize the generated power and minimize the standard deviation of the joint power output. To address the nonlinearities, multi-objective optimization, and operational decisions, a unified framework for the complementary operation of the world's largest hydro-PV plant is established [179]. Matlab Simulink Design Optimization (SDO) and HOMER were used to analyze the performance of an integrated PV/HKT/DG/Battery system. The results showed that the SDO hybrid systems are the best choice for scaling because of their larger batteries with smaller photovoltaic panels and obtain the least amount of available energy contained in batteries and the least amount of energy generated by the DG than the second approach with HOMER [180]. The Bees algorithm was used to address the optimum size of grid-independent hydro, solar, wind, and fuel cell hybridization, and the obtained results were compared with PSO. The obtained results illustrate that for the considered load, a solar system is not feasible due to the high capital cost of solar panels, while the wind turbine/hydro/fuel cell hybrid system is the optimally economical design [181]. Municipal solid waste is usually burned for power generation through several processes using waste-to-energy technology. But usually more energy loss in the boiler and exhaust gas will occur during the process, and as a result, the energy efficiency of waste-to-energy technology is only about 20%. To improve the thermodynamic and economic performance of waste-to-energy plants, a novel WHR system comprised of RC, ORC, and ARC has been designed. When compared to

Table 6
Comparison of various artificial intelligence techniques applied to optimize HRES [139–172].

Proposed method	Compared methods	System Components	Location	SA/GC	Objective function	Findings	Economics Obtained Using Various Proposed Techniques	Ref.
ALO, GWA	CS, FPA	PV/WT/SB/GT	Egypt	GC	Total annual cost and System emissions	Proposed ALO and GWA techniques give more reliable results with superiority than the CS and FPA	WT-PV-SB-GT-Utility Grid (Winter Scenario): ALO / GWO / CS / FPA — TAC ($/year): 540,880 / 550,013 / 559,640 / 573,050; Emission (ton/yr)(10^{10}): 1.66 / 1.62 / 1.66 / 1.71. WT-PV-SB-GT-Utility Grid (Summer Scenario): TAC ($/year): 1,549,031 / 1,550,814 / 1,555,800 / 1,555,377; Emission (ton/yr) (10^{11}): 1.62 / 1.57 / 1.65 / 1.75	[139]
GWA	PSO	PV/WT/SB	Ahvaz, Iran	SA	Minimize TNPC	The GWO algorithm was superior and capable of yielding lower cost and higher reliability than the PSO	GWO / PSO — TNPC (M$): 1.199 / 1.201; LPP (%): 0.653 / 0.655	[140]
GWA	DHS, B&B	PV/WT/BT	-	SA	Minimize system total annual cost	GWA is a good candidate, it is Superior to the DHS and B&B. GWA has fast convergence than DHS. B & B does not guarantee a global optimum solution	GWO / DHS / B&B — TAC($): 5470.37 / 5652.66 / 5652.63	[141]
GWA	HS, PSO	PV/BMG/BG/BT	Haryana, India/ To electrify Households	SA/GC	Minimize system NPC	GWO provides more optimized results compared to HS and PSO. GWO converges faster than the compared methods. The optimal configuration in the study region is the grid-connected configuration.	GWO / HS / PSO — NPC($): 636,923.07 / 641,538.46 / 836,923.07; COE($/KWh): 0.088 / 0.089 / 0.099	[142]
GWA	GA, SA	PV/WT/BM	Karabuk University Campus	SA	Minimize TNPC, LCOE	GWA technique has achieved better output performance than the GA and SA.	GWO / GA / SA — LPSP < 0.01: TNPC ($): 697,657 / 728,257 / 763,371; LCOE($/KWh): 0.3061 / 0.3195 / 0.3349. LPSP < 0.02: TNPC ($): 581,428 / 582,685 / 586,714; LCOE($/KWh): 0.2551 / 0.2557 / 0.2574	[143]
GWO	PSO	PV/DG/SB	Djanet, south Algeria / Rural village	SA	Minimize TAC and determine optimal sizing of the system	The proposed optimizer performs fast convergence, lower cost, and higher superior capabilities than PSO	GWO / PSO — ACS ($): 16825,64 / 16872,66	[144]
ABC	PSO, HOMER	PV/WT/BM/DG	Patiala, India/ Small village	SA	To deduce the optimal size of components with the least LCOE by minimizing NPC	The performance of ABC is satisfactory as compared to PSO in terms of computational time and results. Both ABC and PSO perform better than HOMER. ABC provides an optimal combination with LCOE and may be chosen as the best optimizer. ABC shows a minimum deviation than PSO	HOMER / PSO / ABC — NPC($): 7,52,118 / 7,30,011 / 7,23,378; COE($/KWh): 0.181 / 0.176 / 0.173	[145]
IAEO	AEO, PSO, SSA, GWA	PV/WT/FC	Ataka, Egypt	GC	Minimize COE	IAEO provides fast convergence characteristics and minimum COE. When compared to other techniques proposed, IAEO could be used to solve complex and non-linear problems.	PV/WT/FC — IAEO / AEO / PSO / SSA / GWO: NPC($): 9,957,537 / 9,972,230 / 10,210,924 / 10,078,019 / 10,567,013; COE($/KWh): 0.41243 / 0.41304 / 0.422926 / 0.417422 / 0.437675. WT/FC: NPC($): 10,865,694 / 10,870,047 / 12,784,625 / 11,285,398 / 11,055,336; COE($/KWh): 0.45004 / 0.45023 / 0.529527 / 0.467430 / 0.457901. PV/FC: NPC($): 9,423,136 / 9,448,473 / 10,044,733 / 9,638,349 / 10,007,762; COE($/KWh): 0.39029 / 0.39135 / 0.416043 / 0.39921 / 0.414512	[146]
ICSA	CSA, PSO	PV/WT/SB	Zanjan, Iran / Residential load	SA	Minimize TNPCS	More convergence speed. Less convergence tolerence. Low TNPCS. Better reliability	PV/Batt (DPHIPmax=3%): ICSA / CSA / PSO — NPCS(M$): 1.050 / 1.053 / 1.051. PV/WT/Batt (DPHIPmax=3%): NPCS(M$): 1.052 / 1.061 / 1.054. WT/Batt (DPHIPmax=3%): NPCS(M$): 1.398 / 1.406 / 1.401	[44]
CS	PSO	PV/WT/SB	Ghardaia, Algeria / Residential & cattle farm load	GC	Minimize TCOS. Minimize grid utility	Needs fewer iterations to converge the best configuration. Faster convergence. Better accuracy	Residential / Farm — Optimal system cost ($): 15,705 (GPAP 0.25%) / 45,790 (GPAP 2.5%)	[52]
CS	GA, PSO	PV/WT/SB	Almora / India	SA	To satisfy demand reliably, Minimize total system cost	Faster in optimization. Reduces computational burden. Minimal computer resource utilization. Fine-tune parameters are less	Optimized cost($): PV/SB — CS / PSO / GA: 2912505 / 2921016 / 2921661. WT/SB: 885803 / 920091 / 897451. PV/WT/SB: 768950 / 772575 / 810460	[45]
FA	APSO, GEWA, BA	PV/WT/SB	Bouzaréah, Algeria /Household	SA	Minimize electricity cost	Convergence optimal solution more quickly. High accuracy. Effective in dealing with multimodal and nonlinear problems	FA / APSO / GEWA / BAT — PV/wind/battery (1% LDR): EC ($/kWh): 1.1666 / 1.1666 / 1.1685 / 1.1780. PV/battery (1% LDR): EC ($/kWh): 1.1903 / 1.1903 / 1.1903 / 1.1990. Wind/battery (1% LDR): EC ($/kWh): 6.8504 / 6.8505 / 6.8629 / 6.8505	[46]
CS, FA	PSO	PV/WT/SB		GC	Minimize Total annual cost. To increase the reliability. Reduce the effect of pollutants	Proposed techniques are better, more powerful, and recommended for use to get the optimal solution for hybrid power generation systems.	CSA / FA / PSO — WT-PV-SB (Winter Scenario): TAC ($/year): 1 / 1 / 0.9579; Emission (ton/year): 0 / 0 / 17.0836. WT-PV-SB (Summer Scenario): TAC ($/year): 1 / 1 / 0.9521; Emission (ton/year): 0 / 0 / 59.8295	[47]
FPA	TLBO, PSO	PV/WT/FC/HS	Ardabil, Iran	SA	Minimize TNPC	Easily implement and determine optimal decision variables. Fast convergence. Lower cost. Better reliability values for different indices	FPA / PSO / TLBO — PV/FC (5% $LOEE_{Max}$ and $LOLE_{Max}$): TNPC (MUS$): 3.1415 / 3.2617 / 3.8453. PV/WT/FC (5% $LOEE_{Max}$ and $LOLE_{Max}$): TNPC (MUS$): 2.6724 / 2.8145 / 2.9802. WT/FC (5% $LOEE_{Max}$ and $LOLE_{Max}$): TNPC (MUS$): 5.6518 / 5.7164 / 5.81625	[147]
Hybrid IHS-SA-GIS	HS-GIS SA-GIS	PV/DG/SB	South Khorasan, Iran	SA	To minimize TLCC, LCOE	The hybrid algorithm obtained more accurate results compared to HS and SA. Probability of finding the global solution for a complex optimization problem increases.	IHS-SA-GIS / HS-GIS / SA-GIS — Solar-diesel-battery: TLCC ($): 17900 / 18360 / 17900; LCOE ($/kWh): 0.4429 / 0.4571 / 0.4441	[41]
MFO, MVO	FPA	PV/WTSB/FCGT	Egypt	SA/GC	To minimize total Annual cost, System pollution	Proposed MFO and MVO are more superiority and effectiveness than using FPA	MFO / MVO / FPA — Winter Scenario: SA / GC / SA / GC / SA / GC: TAC ($/year): 279,486 / 376,851 / 253,246 / 342,235 / 294,320 / 382,419. Summer Scenario: TAC ($/year): 466,041 / 563401 / 427,671 / 524,536 / 489,612 / 572,098	[148]
PSOGSA	PSO, GSA	PV/WT/DG/SB	Pamarawan Island, Philippines		To minimize LCOE, LPSP	PSOGSA gained the lowest cost of electricity followed by GSA and PSO. PSOGSA provides a better record benchmark	PSOGSA / PSO / GSA — LCOE($/KWh): 0.23176 / 0.26608 / 0.25537; LPSP (%): 0.10906 / 0.11109 / 0.10507	[149]
SSO	HHO, GWO, MVO, ALO and WOA	PV/WT/DG/SB	Sakaka, Saudi Arabia	SA	To minimize COE	Simple construction. Require less controlling parameters. Superior for reliable design and costless MG. Consumes less elapsed time	SSO / WOA / ALO / MVO / GWO / HHO — COE($/KWh): 0.1349 / 0.2523 / 0.1438 / 0.2326 / 0.1587 / 0.3780	[150]

TLBO	GA, PSO	PV/SB	Tehran, Iran	GC	To Minimize NPC, COE	TLBO's superior in solving optimization problems than GA and PSO Reaches final solution faster than PSO and GA Provides best convergence rate Simple execution	TLBO PSO GA TNPC ($) 23978 23985 23990 LCOE($/KWh) 0.174 0.175 0.176	[151]
WCA, MFO	GA, FA, GWO	PV/BG/PHES/SB	Silchar, India/ Radio transmitter station	SA	To minimize TNPC	Both WCA, MFO are effective to determine the optimal system. WCA provides a better solution than other algorithms WCA convergence faster than MFO and avoids traps at local optima	WCA MFO GA FA GWO PV-BG-PHES-SB (LLP=0) LCOE($/KWh) 0.4864 0.4865 0.9632 0.9116 0.9109	[152]
DHS	DSA	PV/WT/DG/SB	Rafsanjan, Iran	SA	To minimize total annual cost and environmental impacts	DHS outperforms DSA in terms of the accuracy DHS is easy to implement DHS leads to obtaining the system's objective function	DHS DSA PV/WT/DG/SB TAC ($) 15790.38 15843.67 PV/DG/SB TAC ($) 19963.8 19963.8 WT/DG/SB TAC ($) 15672.94 16036.8 DG only TAC ($) 64055.46 64055.46	[153]
GA	HOMER	PV/WT/BG/BM/FC/SB	Chamarajanagar, India	SA	To minimize TNPC, COE, To reduce Unmet load, CO_2 emissions	By comparing GA and HOMER optimization techniques, GA is the more cost-effective technique with the least NPC, more PV penetration, and fewer CO_2 emissions.	GA HOMER . PV-WT-BG-BM-FC-SB COE ($/kWh) 0.163 0.214 NPC ($) 856013 890013	[154]
PSO, IWO, BSA	Hybrid IWO-PSA	PV/WT/SB	Rabat, Morocco/Irrigation unit	SA	To determine the optimal size of components that minimize TNPC	Hybrid IWO-PSO predicts the least system NPC with a minimum COE of 0.023/kWh Compared to other algorithms, IWO-PSO is the most efficient in terms of speed of convergence and results.	PSO, IWO BSA IWO-PSO NPC ($) 52513.962 54037.255 49656.7544	[155]
WOA, WCA, MFO	PSOGSA	PV/WT/DG/SB	Abu-Monqar, Egypt	SA	To determine the best optimal size based on COE	The WOA technique is capable of obtaining the minimum value of an objective function In terms of economic and convergence efficiency, WOA operates at a high level	WOA WCA MFO PSOGSA NPC ($) 5418863.799 5869088.725 5869036.077 6030997.882 COE($/KWh) 0.1850043 0.2003754 0.2003736 0.205903	[48]
HBB-BC	PSO, DHS	PV/WT/SB	Qazvin, Iran	SA	To minimize TNPC	Hybrid HBB-BC indicates the best performance with high accuracy compared to PSO and DHS. Compared to DHS, PSO is better.	HBB–BC PSO DHS PV/wind/battery (2% ENS_{Max}) TPC ($) 1,794,835 1,794,867 1,805,291 PV/wind/battery (5% ENS_{Max}) TPC ($) 1,468,926 1,468,944 1,478,610 PV/wind/battery (10% ENS_{Max}) TPC ($) 1,159,481 1,159,485 1,166,839	[156]
HS	GA, PSO	PV/BM	Bardsir, Iran	SA	To minimize TNPC	HS indicates the best performance than GA and PSO While comparing GA and PSO, GA is outperformed by PSO	HS GA PSO PV/Biomass TNPC ($) 1672296 1677406 1672705 LCOE ($/kWh) 0.1855 0.1860 0.1856	[157]
CSA	GA, PSO	PV/WT/Tidal/SB	-	SA	To minimize TNPC	CSA finds the most accurate outcomes, requires less run time to find an optimal solution and has a better convergence rate than GA and PSO.	CSA PSO GA TNPC ($) 1271407.27 1300061.09 1473506.54	[158]
MBA	PSO, CS, ABC	PV/WT/FC	Helwan, Egypt	SA	To achieve minimum system annual cost	The Proposed MBA technique saves costs about 24.8%, 11.5%, and 8.9% corresponding to PSO, ABC, and CS algorithms respectively. MBA requires less time to reach an optimal solution.	PSO CS ABC MBA PV/WT/FC Total cost ($) 5598167.26 4623953.2103 4759967.5368 4209827.3116	[159]
ACO_R	GA, ABC, B&B	PV/WT	-	SA	To minimize the total design cost	ACOR outperforms other techniques in terms of optimal results and speed. The average optimal convergence and times of the proposed technique are lower than the other algorithms	ACO_R ABC GA TNPC ($) 6692.6132 6692.6132 6692.6132 Convergence time 0.44381 0.68183 0.55029	[160]
GA-PSO, MOPSO.	HOMER	PV/WT/SB	Tehran, Iran	SA	To minimize the total present cost To maximize reliability	MOPSO gives a wider choice of optimal points, which optimizes two objectives simultaneously Compared to the HOMER single objective technique, GA-PSO has more convergence rate and better results.	GA-PSO HOMER COE($/KWh) 0.508 0511 NPC($) 787193 1652300	[161]
Hybrid HS-SAA	HS, SA	PV/WT/Biodiesel/SB	Fedeshk, Iran	SA	To minimize life cycle cost To reduce Emissions	Hybrid HS-SAA improves the solution through better quality and fast convergence rate compared to individual HS and SA	HHSSAA SAA HSA PV/BDG/BA LCC($) 20,534 20,534 20,674 CPU (Sec) 1872.511 1978.826 1887.75	[162]
CLONALG	GA	PV/WT/SB	Troyes Barbery Station, France Qena Al-Gadida city in Egypt	SA	To determine the optimal systems with minimum cost and fewer fluctuations in output	The proposed method is very effective for selecting a suitable design and achieving its objectives. Compared to GA, the proposed one provides better power reliability and requires less battery capacity.	CLONALG Troyes Barbery Station Cost ($) 1237855.095 Qena AL-Gadida Cost ($) 741039.85	[163]
FPA	ABC, PSO	PV/FC	Egypt	SA	To formulate minimum Total annual cost To maximize reliability	The proposed FPA technique has less fulfillment time, better rendering, and superior performance when compared to ABC and PSO	FPA TNPV ($) 3,244,897 LCOE($/KWh) 0.334	[164]
IBA	HSA	PV/WT/SB/FC/HSS/ROD	South Khorasan, Iran	SA	To determine the most cost effective system design To minimize total LCC cost To maximize LPSP	The performance of the IBA is better than the HS algorithm Battery energy storage is more cost-effective than hydrogen storage	IBA HSA PV/HSS/RO (Optimal system for LPSP 2%) Total cost ($) 2,730,760 2,778,680 PV/BS/RO (Optimal system for LPSP 2%) Total cost ($) 238,580 445,050	[165]
MOCSA	MOPSO	PV/DG	Kerman, Iran	SA	To minimize TNPC, LPSP, and CO2 Emissions	MO-CSA provides more promising results than MOPSO as the proposed algorithm obtains well-distributed and widely spread Pareto front The cost of the system increases significantly when the operating reserve is considered	MO-CSA (For LPSP 0%) With Operating reserve Without operating reserve TNPC ($) 4.016×10^5 3.727×10^5 $TECO_2$ (kg) 2.527×10^5 2.322×10^5	[166]
Jaya, HS	PSO	PV/WT/BM/SB	Lat. 29° 49.794' N Lon.39° 34.362' E	SA	To determine the minimum value of effective NPC, high efficiency, and reliability system,	The HS technique has faster convergence, and minimum deviation and is more efficient at determining the optimal system than the Jaya and PSO algorithms.	HS PSO Jaya PV/Bio/WT/Battery NPC($) 581,218 591,402 590,137 COE($/KWh) 0.254 0.259 0.258	[50]
MFFA	FFA	PV/WT/FC	Ataka, Egypt	GC	To minimize the COE To ensure high reliable power supply with low energy fluctuations within the utility grid	MFFA reaches the best objective function faster with fewer iterations and implementing time than FFA	FFA MFFA COE($/KWh) 0.3570 0.3570 No.of iterations 11 8 Execution time 6792.7 3664.6	[167]
FA	SFLA, PSO	PV/WT/FC	Beni-Suef, Egypt	SA	To design the optimal system with minimum Levelized cost of energy	The Effectiveness of FA is better than SFLA and PSO algorithms	FA PSO SFLA PV/WT/FC COE($/KWh) 0.45 0.51 0.45 NPC ($) 2,698,325 3,073,583 2,698,325 Executing time (S) 118.61 131.00 909.88	[168]
HFA/HS	PSO	PV/WT/FC	Hurghada, Egypt	GC	To determine the optimal sizing at the lowest COE	The proposed HFA/HS algorithm takes much less execution time than the PSO algorithm to find the optimal solution and shows better performance Using FC provides better performance than battery banks	HFA/HS PSO COE($/KWh) 0.887 0.888 NPC ($) 8,312,900 8,317,000	[169]

BBO	GA, PSO	PV/ BG/ BM/ MHG/SB	Uttarakhand, India	SA	To develop an optimal system with minimum COE	CC strategy is more cost-effective than LF and PSS The cost of energy obtained using BBO is the most cost-effective compared to PSO and GA algorithms. The cost of energy decreases significantly when the clean development mechanism is considered	BBO / PSO / GA PV/ BG/ BM/ MHG/SB (Optimal system) COE(₹/KWh) 5.64 / 5.66 / 5.67 NPC (₹) 33,076,784 / 33,208,069 / 33,309,626	[53]
GA	PSO	PV/WT/BM	Barwani, India	SA	To minimize electricity price	Less DG operating hours, minimum COE, less algorithm running time, and also reliability is higher for cycle charging strategies using PSO	GA / PSO Cycle charging COE ($kWh) 0.2396 / 0.2393 Load following COE ($kWh) 0.2625 / 0.2617	[170]
TLBO	BFSO,GA,PSO	PV/WT/BM/SB/VDG	Barwani, India	SA	To minimize COE, LPSP, PMI To maximize RF, HDI, JCI	For six different combinations of hybrid systems, the best solutions are obtained for TLBO, then BFPSO, PSO, and GA For all objectives, TLBO has superior performance in terms of technical, economic, and social parameters	GA / PSO / BFPSO / TLBO PV/WT/BM/DG/SB (Optimal system) COE($/KWh) 0.205789 / 0.196498 / 0.196945 / 0.195166	[42]
Hybrid TLBO-CS	GA, PSO	PV/WT/DG/SB	Jeju Island, Korea	SA	To minimize TAC, LPSP	The proposed Hybrid TLBO-CS can perform exploration and exploitation simultaneously and is more efficient than GA, PSO	GA / PSO / TLBO-CS PV/WT/DG/SB (LPSP < 4%) Total cost ($) 8.83×10^4 / 8.87×10^4 / 8.82×10^4	[171]
PSO, TS, and SA	IPSO, HIS, IHSBSA, ABSO	PV/WT/SB	Rafsanjan, Iran	SA	To minimize TAC	ABSO yields more promising results quickly than IHSBSA, IHS, IPSO, PSO, TS, and SA in terms of the TAC	ABSO / SA PV/WT/battery (For LPSP 2%) Total cost ($) 56281.59 / 106140.5 Rank 1 / 7	[172]

traditional WTE plants, the proposed system increased energy and exergy efficiency by 37.66% and 35.65%, respectively. By using butane as a working fuel, more electricity is generated, and from an economic point of view, investing in the proposed waste-to-energy technology is worth it and environmentally beneficial [182]. A two-layer nesting model was developed to optimize the PV capacity of small and medium-sized hydro-PV systems under different scenarios, considering the uncertainties of runoff from hydropower plants and PV power output. This framework was applied to the Naijili hydro-PV hybrid system, and it investigates PV capacity under different levels of confidence, the effects of global hydropower expansion on solar capacity, and predicted power generation. The result of this study shows that the optimal capacity of PV lies between 2 MW and 14 MW under different scenarios, and at an 80% confidence level of PV capacity, the hydropower plants could stabilize the total system power output. From the obtained results, it has been proved that the proposed model and the solution method are more practical in determining the optimal PV capacity [183].

Loss of diversity and premature convergence will occur when WOA is applied to large-dimension optimization problems. To overcome the demerits of WOA, a modified NSWOA is developed with improved population diversity at the last stage of evolution. The modified NSWOA was proposed to effectively optimize hybrid hydro-PV-wind power generation to maximize total power generation and mitigate power output fluctuations. The hydropower plant overcomes the inherent variability of PV and wind power and stabilizes the hybrid power-generating system [184]. Based on the complementary guarantee rate and cumulative time proportion, Universal frameworks were explored to optimize the capacity of a grid-connected large-scale hydro-PV-wind system to achieve the minimum cumulative fluctuations. The specific power output characteristics of solar and wind, time proportions, hydropower station characteristics such as turbine units, reservoir regulation, runoff characteristics, and hydraulic connections are considered to determine the optimal capacity of PV and wind turbines. To achieve energy conservation and clean, emission-free generation, this study will provide technical support for the development of integrating hydro-PV-wind hybrid systems [185]. Three short-term operating techniques for the WSH hybrid device have been established, and the simulation estimation method is being considered to investigate uncertainties in wind speed, solar radiation, and power load at the same time. The primary objectives are to optimize smoothing wind and solar performance, reduce the root mean square error (RMSE) between the time output and load curves, which facilitates the integration of the WSH system into the power grid, and maximize overall power generation [4]. For the application of scheduling of short-term hydropower reservoirs IA-PSO algorithm has been applied to attain maximum power generation from hydro. For this, the Geheyan reservoir and the Gaobazhou reservoir in the Qingjiang River basin, Hubei province were considered. The efficiency and robustness of IA-PSO have been compared with the conventional operation method, dynamic programming, and the conventional PSO algorithm. The proposed algorithm is more accurate and precise, and cascading reservoirs enable 4.83% more electricity than conventional operation [186]. While scheduling short-term hydro, consideration of a risk measure is important to maximize the profit of the hydropower producer to survive in a competitive framework and minimize risk. A mixed-integer nonlinear programming approach is applied to attain the maximum total profit for the hydropower producer while considering risk management. For modeling and simulations, a cascading hydroelectric system in Portugal has been considered. Moreover, the effects of the head, start-up costs, multiple operating regions, and various hydro constraints were simultaneously taken into account to solve a cascading hydro system [187]. From an environmental and modeling viewpoint, a study of present and prospective environmental constraints on the scheduling of hydro-dominated structures has been conducted [188]. The optimum generating scheduling of hydropower plants should incorporate many electrical and operational constraints into account, including the hydro plants' dynamic hydraulic coupling, water discharge, and storage limits, hydro unit power output, the system's power load requirement, and the hydro plants' water balance [189]. The scientific literature has reported the major contributions of the different techniques for each specific hydro system configuration [190]. A vast variety of variables and constraints determine the optimal working of hydroelectric systems, reservoirs, and associated power stations. Since the objective functions of hydropower systems are nonlinear, it is extremely complicated to operate and optimize. Depending on the size of the hydro scheme, the available computing power, and the underlying electricity demand, the hydro producer requires appropriate scheduling tools. As a result of the increase in hydro-related multi-energy production becoming a high-priority research area, this will be a popular research topic in the future. Complementary energy systems are also being studied extensively, with increased emphasis on scheduling and optimization to ensure reliable hydro system operation. Recently, numerous artificial intelligence optimization approaches that can provide effective solutions to multi-objective, non-linear, real-world complex problems have been developed. Therefore, it is vital to apply new stochastic optimization methods to overcome the challenges faced by hydro-hybrids.

GWO is a swarm-based optimization algorithm inspired by the hunting process and leadership hierarchy of the grey wolf. Indeed, GWO has a simple structure and straight forward to implement and has strong exploitation ability [191]. It can be noticed that, after the year 2015, researchers have a huge interest in applying GWO, which provides better quality results than GA and PSO. However, it suffers from low solving accuracy for local optima and a low convergence rate. The Ant optimizer is inspired by the foraging behavior of antlions, which follows several steps while hunting prey, including "random walks of ants, building traps, entrapment of ants in traps, catching prey, and re-building traps." It has several merits such as being simple, scalable,

flexible, local optima avoidance, and providing a better balance between the exploration and exploitation phases [192]. During the years 2015–2019, numerous researchers used the Ant Lion optimizer for different applications, and the total number of papers published during this period was 633. To improve the performance of ALO, Several enhancements have been done to ALO and proposed various modified versions of ALO by various reachers such as Binary ALO, binary coded of ALO, Enhanced ALO, chaotic ALO, etc. However, particularly when using ALO to tackle problems in real-world applications, parameter tuning is crucial. ALO should be improved since one of its key shortcomings is that it cannot manage a large number of variables, and currently, no work has been done to handle such a large number of variables [193]. Cuckoo search is one of the most effective optimization techniques, which was inspired by some cuckoo species' obligate brood parasitism. During the optimization process, it requires less parameter adjustment compared to GA and PSO. Due to this it can able to ensure the global optimum results without trapping into local minima and becomes more potential and superior to GA and PSO. To initialize the nest location cuckoo search algorithm uses a random number and due to this sometimes the nest location may be the same or not dispersed properly which makes a chance to provide a locally optimal solution. Also, it requires changing its parameters' values with the iterator process, suffers from boundary issues, and levy flight, which affects its convergence rate, and cannot provide better performances for discrete and multi-objective problems [52,194]. The crow search algorithm was inspired by the intelligent behavior of crows, which hide their excess food in a place and, whenever required, retrieve the place. Also, due to their greedy behavior, the crows can find and steal the hidden food of other crows. CSA has gained considerable interest in determining the global optimal solutions to optimization problems due to their simplicity, fewer parameters, and greater flexibility; hence for the optimal design of renewable-based microgrids CSA provides more efficient outcomes than GOA and HHO. However, to obtain a global solution, CSA requires a well-balanced balance between the exploration and exploitation phases throughout their search process [195–197]. The flower pollination algorithm is inspired by the pollination behavior of flowers, which involves cross and self-pollination, flower constancy, and global and local pollination. Its impressive characteristics have attracted many researchers to employ it in the optimal sizing of renewable-based design problems. It requires fewer parameters and is flexible and simple to implement. FPA requires less time to determine a convergent optimal solution when compared to PSO, and ABC. However, it requires proper modification to solve discrete optimization problems. The lack of a perfect balance between global exploration and local exploitation is one of the shortcomings that confine its applications to optimization problems. Hence, FPA requires various strategies to enhance its searchability [164,198,199].The water cycle algorithm was inspired by the natural water cycle and developed using observations of how rivers and streams naturally flow toward the sea [200]. The biological activity of moths fighting flames in nature inspired the moth-flame optimization algorithm [201]. The WCA and MFO algorithms have a higher rate of convergence to the optimal outcome than GA methods. In contrast to MFO, WCA offers a somewhat superior design outcome. The minimal objective functions achieved using the current techniques are significantly lower than those produced using GA, FA, and GWO-based findings. While comparing MFO, the convergence rate of WCA to the optimal solution is better. The WCA and MFO algorithms outperformed some of the other algorithms previously employed in developing hybrid systems, according to a comparison of the performance of the most optimal configuration with designs from the literature [152]. WCA may fail to achieve an appropriate balance between the exploitation and exploratory phases and get stuck in the local optimum, resulting in premature convergence for complicated problems [202]. Premature convergence and poor population diversity are drawbacks for MFO [203]. The social spider algorithm was created to address global optimization issues and is based on the foraging method used by social spiders. This algorithm uses vibrations on the spider web to detect the positions of prey [204]. When compared to HHO, GWO, MVO, ALO, and WOA, the SSO is superior in designing reliable microgrids that are also economical [150]. The SSO algorithm typically has some drawbacks, including slow convergence rates, poor calculation accuracy, and a low success rate for high-dimensional functions, as well as a propensity to easily enter local optimization and a contradiction between population diversity and the rate of convergence [205]. Among all the discussed algorithms, GWO is determined to be most frequently used for hybrid system sizing among other algorithms, although having certain drawbacks it can be found GWO performs better than GA and PSO. Also in recent years, various algorithms such as CS, CSA, FA, HHO, MVO, SSO, WOA, MFO, and WCA have been applied to hybrid system optimization studies, and among these algorithms, SSO has provided superior performance. Not every optimization algorithm and its variations provide the most optimal solutions to a given problem. Additionally, even though some optimization approaches are effective, these techniques need to be improved to increase their performance. To improve the discipline of artificial intelligence based optimization techniques, new optimization strategies must continuously be constructed because it is still quite difficult to find ways to speed up an algorithm's convergence.

5. Hybrid optimization techniques applied to optimal sizing of HRES

Hybridization is a concept used in the field of engineering systems analysis to describe the method of integrating various optimization methods into a single approach. The combination of any different approaches has a synergistic effect, which adds a significant aspect by enhancing or emphasizing the attributes of both strategies for solving complex optimization problems. Hybrid optimization algorithms result in a better outcome and outperform single algorithms in terms of efficiency. Numerous preliminary studies show that hybrid algorithms are designed to get as close to the global optimum points as possible with the least amount of iteration. Numerous researchers are currently advocating the expansion of existing single algorithms into new hybrid methods, which promise to vastly improve exploitation and exploration capacities [206]. The authors have developed LQE models of the proposed hybrid gravitational search algorithm and teaching-learning-based optimization algorithms for future EDE studies. In the first step, traditional GSA calculates the best solution in the global search space. A new search space value is determined based on the GSA's best solution, and the next step is to reach the best solution in the local search space using the standard TLBO algorithm [207]. Because there is no inertia weight to regulate particle speeds and no satisfactory balance formed between local and global searches, the SSO solution performs poorly in a variety of functions. To overcome these problems, the authors proposed the FASSO technique and applied it to solve the simultaneous computation of various benchmark test functions that have special conditions and features [208]. The PSO and SA are hybridized to further improve the social behavior of a swarm, and their performance has been evaluated by different dimensions of benchmark functions for global minimization [209]. Harmony search has poor performance in global search, and, in certain cases, its convergence rate can also be a problem. MBA is a memory-less optimizer and suffers from low local search precision. To tackle these drawbacks, a new hybrid mine blast harmony search has been developed by embedding HS operators into MBA, which improves the diversity of the population in HS and enhances the exploitation capabilities in MBA [210]. Recently, several hybrid techniques have been developed and applied to a huge variety of applications with great success. In the following, some new hybrid techniques in this area are described. By updating the hierarchy and position of GWO and fusing the strengths of GWO and SFLA, a new hybrid algorithm named SGWO has been proposed. Further, SGWO has been improved with the strategy of differential evolution as a new SGWOD hybrid. These algorithms have been applied to solve different

benchmark functions, which show promising performance in exploration and exploitation, but the processing time of these algorithms is higher and requires improvements to their efficiency. Also, these algorithms are implemented to predict the daily load on the power network [211]. Many researchers have hybridized genetic algorithms with other algorithms to overcome the limitations of GA. A population-based stochastic optimization technique that combines binary-coded GA with a self-organizing migrating algorithm is the self-organizing migrating genetic algorithm (SOMGA). SOMGA, on the other hand, has not been widely applied to the optimization problem associated with increased search space and high precession. To overcome the shortcoming of SOMGA, a new hybrid algorithm called RCSOMGA, which fuses the features of RCGA and SOMA, was developed. The proposed hybrid RCSOMGA optimizer is applied to solve nonlinear bound-constrained optimization problems, and while comparing with other algorithms, this algorithm provides a better solution in terms of various performance characteristics with less computation effort [212]. The authors proposed a hybrid algorithm by integrating the grey wolf optimizer and the crow search algorithm that combines their respective strengths and has been employed to solve unconstrained function and feature selection problems [213]. To solve the demerits of PSO, such as local minimum and debility, the position updating formulae in SCA and the levied flight approach are used, and hence the exploration performance of PSO has been enhanced. Therefore new hybrid PSO based on sine, cosine, and the levy factor (PSOSCALF) was developed and applied to different benchmarks and constrained problems. The results show that the proposed method has a higher convergence rate and is more superior and efficient than the other algorithms presented [214]. Nowadays, integrating the merits of any two optimization algorithms has attracted more interest in developing hybrid optimization algorithms. The authors developed three hybrid techniques based on seagull optimization, such as SOA-TEO1, SOA-TEO2, and SOA-TEO3, and applied them to improve the accuracy of feature selection problems. In the SOA-TEO1 fusion method, to obtain a proper balance between the exploitation and exploration phases, in the iteration process, a location update formula has been set to choose an algorithm randomly. Whereas in SOA-TEO2 fusion, for location iteration, the SOA algorithm has been used, and for the final location update, the TEO algorithm has been used. In the hybrid SOA-TEO3, to improve the SOA exploitation phase, the heat exchange idea of the TEO algorithm is applied. While observing the results of three fusion methods, SOA-TEO3 is superior, provides better results, and has the best stability [215]. Hybridize krill herd annealing with a quantum particle swarm optimization algorithm and propose a new artificial intelligence, the AKQPSO algorithm, which combines the better performance of QPSO exploitation and AKH exploration. The shortcomings of the KH technique's poor local search ability are optimized using the SA strategy, and PSO is optimized by introducing quantum behavior, which improves the searchability of the proposed algorithm with better performance. The teaching–learning-based algorithm (TLBA) is one of the most recent artificial intelligence strategies that are being used to propose the best solution method for different research problems. In previous research, several hybrid techniques have been studied and implemented for optimally designing hybrid renewable energy systems [216].

A hybrid GPSBBO optimization technique that combines the exploitation capability of biogeography-based optimization (BBO) with the exploration capability of particle swarm optimization (PSO) is used to optimize the overall system cost and reliability index of a hybrid wind-photovoltaic energy system [217]. An economic analysis for the PV-WT hybrid energy system has been conducted by considering with and without ESS during grid interruption and curtailment. Existing PV and wind farm generation is combined as a hybrid system, formed into a 22 KV feeder, and connected to a 110/22 KV substation in the Indian state of Tamil Nadu. For economic analysis, the RBFNOEHO hybrid technique is utilized, in which initially the amount of load demand for HRES is predicted by RBFNN, and then the OEHO technique is used for optimal system identification [218]. A multi-objective hybrid grey wolf optimizer is employed for the complementary scheduling model of hydro-thermal renewable power systems. The findings indicate that integrating the operation of renewable energy and hydropower plants favors both economics and pollution, with operational costs and pollutant emissions falling by 11.9% and 17.4%, respectively, when a hybrid system's RE capacity is raised from 50% to 100% [219]. The MOHTLBOGWO hybrid algorithm is proposed to determine the optimal placement and sizing of the PV and wind systems in the distribution network (33 and 69 bus). This algorithm is implemented in the form of single- and multi-objective optimization with the aim of loss reduction and improving system reliability. The dispersion of photovoltaic and wind turbine sizes would lead to a greater reduction of losses and their associated costs, more reliability improvement, and better net savings on the network [220]. As the generation sources are highly governed by environmental variations, there is a need to effectively identify the location of the generation systems to meet the load requirements with cost-effective and reduced line losses to improve the voltage profile. A robust hybrid BAPSO algorithm is proposed to identify the optimal placement of the solar generation and optimize the capacity of a solar PV-battery-based microgrid to minimize transmission losses. The bat algorithm, which has the property of triggering control over space exploration, is employed to overcome the shortcomings of PSO, such as premature convergence and intensive computation [221].

For multistep wind speed predictions, three new hybrid models—EMD (LF)-ELM, WPD-EMD (HF)-ELM, and the WPD-EMD-ELM model were employed. The WPD is chosen to decompose the original wind speed series into several sub-layers, while the EMD is utilized to further decompose low frequency, high frequency, and all the sub-layers into several intrinsic mode functions, respectively. Finally, the ELM is used as a predictor with robust generalization and high training speed to process the decomposed sublayers to predict the wind speed. Among the involved models, the WPD-EMD (LF)-ELM has obtained high-quality predictions [222].The potential of the hybrid SVM model with PSO, BAT, and WOA for predicting diffuse solar radiation in Beijing, China, has been employed. To test the model, six different input combinations have been tested, and the results prove that the input combination of sunshine duration and global radiation improves the diffuse solar radiation prediction accuracy. While compared with conventional SVM, hybridized SVM-BAT remarkably enhanced the prediction accuracy with relative decreases in root mean square deviation of daily diffused solar radiation modeling, followed by SVM-WOA and SVM-PSO [223].

Based on fast ensemble empirical mode decomposition, the two hybrid models FEEMD-MEA-MLP and FEEMD-GA-MLP have been utilized for accurate predictions of wind speed. The FEEMD algorithm was employed to decompose the real wind speed data series into several sub-layers to reduce the complexity of the MLP. The MLP neural networks are used to predict the decomposed wind speed sub-layers, and the MEA and GA algorithms are used to select the optimized initial parameters [224]. A hybrid PLF-NSGAII-MOPSO model has been developed to determine the optimal allocation of ESSs, optimal planning of generator output, and the optimal value of bus voltages to achieve the lowest system total cost while accounting for economic constraints and to optimize carbon emissions while accounting for correlation among wind farm power uncertainties [225]. The weakness of the GWO algorithm is poor global exploration. The introduction of an oppositional learning strategy for decision-layer individuals increases the diversity of GWO. A new hybrid GWOCS has been developed to strike a balance between global exploration and local exploitation, and its effectiveness is being assessed using various benchmark functions. Because obtaining the parameters of PV cell models is critical, the proposed GWOCS is used to obtain the photovoltaic parameters under various operating conditions. For parameter extraction problems, four different PV models were used, and the precision, statistical results, and convergence rate of GWOCS outperformed the other methods compared [226]. While modeling solar PV, the generation of an inaccurate current-voltage curve has resulted in

low efficiency.The earlier optimization methods employed do not guarantee the global optimum, and hence the authors have proposed a new hybrid BPFPA algorithm (Fusion of ABC & FPA) to extract the parameters of both the one-diode and two-diode solar PV model. The search ability of the proposed algorithm has been strengthened by employing an elite-based mutation operation strategy in the standard FPA algorithm [227].

To update the designs of PEMFC in the future, it is necessary to effectively identify their parameters. Several models, such as those empirically validated, two dynamic, and real-time models, have been employed, and these methods are based on the physical concept of fuel cells. These methods fail to measure some parameters and do not provide a proper modeling strategy. As a result, an improved model of Elman neural network-based hybrid World Cup optimization and fluid dynamics was developed. Fluid optimization is proposed to estimate the parameters, and then the proposed algorithm has been validated under different operation conditions [228]. The tri-objective economic dispatch has been investigated concurrently on small, medium, and large-scale power structures, taking into account generation cost, emission, and transmission power losses, as well as practical constraints such as multi-fuel operation, prohibited operation zones, and valve-point effect. A robust FBHPSO-DE algorithm tends to be used because it has ease of implementation and is efficient enough to be useful in solving multi-objective economic dispatch problems. The result proves that, for solving both single- and multiobjective economic dispatch problems, the proposed algorithm is potent, stable, and capable of handling the problems irrespective of the optimization problem size and complexity [229]. The optimal placement and sizing of renewable systems can reduce system losses, generation capacities, and life cycle costs while enhancing service quality and reliability. To electrify rural areas, an improved harmony search and geographic information system was used to determine the optimal location and sizing of the grid independent of solar and hydrogen systems based on minimizing life cycle cost and system losses. The results show that the optimal size of the systems seems to be a function of increasing system reliability [230]. A thermal-wind electrical energy system has been created by incorporating an eleven-wind turbine into a standard thermal energy system. Whenever a renewable source is connected to a conventional power system, a new power dispatching strategy and optimization algorithm for both energy systems are required to improve the performance of the hybrid power generation system. To optimize power flow control and power flow analysis in wind-thermal power systems, a hybrid G-TLBO algorithm is proposed to improve the fuel cost and run time of the algorithm, and the authors also suggest the proposed algorithm is suitable for large-scale power generation systems [231]. The authors used simulated annealing based on the -constrained method to analyze various off-grid RBPS with three goals in mind: technical, economical, and environmental. Then in the above work, the hybrid multi-criteria decision-making AHP–TOPSIS technique is applied to rank the set of possible best scenarios for every optimization of configurations, considering different power generations as well as three different energy storage options. Technical criteria, according to the findings of the sensitivity analysis, conflict with economic criteria more than environmental criteria. The limitation of this approach is that the method of assigning weights to all three criteria could be improved [232].

An integrated RE system (solar, DG, and battery) with limited grid supply is designed for rural electrification in Assam, India, to minimize the annual cost of the design. The recently introduced modified multi-strategy fusion artificial bee colony was employed to solve the objective functions, and its feasibility and superiority are compared with HOMER, the original MFABC, and PSO algorithms. The new version of ABC requires less number of control variables than other algorithms. A selection probability rule is proposed to enhance the modified MFABC algorithm's selection criteria. The key benefits of modifying the selection probability are faster convergence, and the ability to obtain a better-quality solution. So it has been observed that the convergence graph of the annual system cost for the proposed algorithm is better because it provides a minimum annual cost and takes fewer minutes to converge than other techniques [233]. The authors developed and implemented a novel optimization approach for real-time power management and control of a DG microgrid system (PV/WT/DG/FG/MT). The ant-lion approach is selected to solve the MOOD problem, which involves three objective functions to minimize, and the weighted sum method is considered to convert the MOOP problem into the SOP. System modeling and simulation were conducted for both stand-alone and grid systems using MATLAB. The results revealed that the grid-connected operating policy is more economical than the standalone connected mode. The achieved results from ALO are compared with those from NBA, PSO, and ABC methods. It can be observed that the proposed algorithm converges to a globally optimal solution with fewer iterations compared with other heuristic approaches [234]. The authors described a novel operation optimization model for a standalone WT/PV/DG/Battery microgrid system. A combined PSO and SA algorithm was developed to minimize the cost of economic, environmental, and battery depreciation as an objective function. The two objectives of economic and environmental cost are not mutually exclusive, but the battery depreciation cost conflicts with both of them. System performances obtained from SAPSO are compared with those obtained from IWPSO, and a grey target decision-making theory based on the entropy weight method is proposed to identify the optimal compromise solution from the obtained Pareto solution set [235]. The potential of solar and wind farms with pumped hydro storage has been investigated in China, and authors considered the curtailment rate of renewable power to carry out a techno-economic analysis. Multi-objective Particle Swarm Optimization is used to determine the trade-off analysis between the bi-objectives of LCOE and LPSP. The Pareto optimally is used in the simulation of multi-objective function and results show there is a significantly higher value of LCOE with lowering the rate of curtailment rate of renewable sources [236]. The authors used an exploratory optimization method called FPA to evaluate the optimal design of a PV/WT/FC system based on hydrogen storage, and the total net present cost of the off-grid system is minimized by simultaneously considering the reliability constraints LOEE and LOLE. The performance analysis of FPA is studied and compared with TLBO and PSO. The results demonstrated that the wind energy system is not economical and can be seen as an auxiliary system to supply load demand in Iran [147]. The summary of hybrid techniques applied to various applications is illustrated in Table 7.

Because of the higher oxygen concentration in biodiesel, the authors reported that a standalone hybrid system (PV-WT-Biodiesel-Battery) is the most popular choice for meeting load demand in the Fedeshk area of South Khorasan (Iran). The design problem's optimization goal is to minimize system life cycle cost, with PV area, the number of batteries, wind turbine swept area and biodiesel generator fuel consumption as decision variables. For these objectives, the authors suggest a hybrid harmony search-simulated annealing technique that combines the merits of the above-mentioned individual algorithms [162]. Fast charging stations for EVs have been designed by considering demand response and a multi-objective optimization model with the objectives of COE and emissions. Then the MOPSO algorithm is used to solve the designed model, and the TOPSIS method selects the optimal capacity of the FEVCS-WPE system applied in Inner Mongolia, China. The system load demand, objectives, and optimization methods are the three main factors affecting the capacity of the system components. By considering the above-mentioned factors, an optimal system capacity and energy scheduling strategy are developed. Stable energy support can be obtained by connecting the charging station to the utility grid, and the performance of the optimization objective has been improved by taking into account the price-based demand response [239]. The authors developed the modified HGSO by inserting SA into conventional HGSO to determine the optimal capacity of integrated solar and BES in IEEE 69-bus RDS. The optimal size of PV and BES at three buses of IEEE 69

Table 7
Summary of hybrid techniques applied to various applications [206,207,216,217,237,238].

Table 7.
Summary of hybrid techniques applied to various applications

Proposed Hybrid algorithm	Existing method	Test set	Findings	Application	Ref
MFO-TLBO	CSA, MFO, TLBO, GWO, WOA	Seven unimodal Six multimodal Eight fixed-dimensional multimodal	Outperforms traditional MFO The convergence and success rate of the hybrid approach dominates Attain superior outcomes for all benchmark functions (Minimization) compared to other algorithms	Visual tracking	[206]
GSA–TLBO	GSA, TLBO, MLR	Linear, Quadratic Exponential, Classical regression	The findings obtained with the proposed hybrid approach are consistent with training data with correlation coefficients. The proposed approach could regulate the strategy for energy imports and thus make energy policies more realistic.	Energy demand estimation	[207]
GPSBBO	DE, GA, PSO, BBO, Blended BBO, Hybrid BBO-DE	Ackley, Fletcher, Griewank Penalty no. 1, Penalty no. 2 Quartic, Rastrigin, Rosenbrock Schwefel 1.2, Schwefel 2.21 Schwefel 2.22, Schwefel 2.26 Sphere	BBO-PSO outperforms traditional BBO, PSO, and blended BBO in all standard benchmark test functions as a function of the fitness function's minimum and mean values. For nine of the thirteen benchmark functions, the BBO-PSO algorithm outperformed both the DE and GA algorithms. For eight of the thirteen benchmark functions, the BBO-PSO algorithm outperformed the BBO-DE algorithm. The BBO-PSO algorithm can see a much higher success rate than traditional PSO, GA, BBO, and B-BBO. However, the success rate of the DE and the BBO-DE is comparable. Simpler than Hybrid BBO-DE	-	[217]
AKQPSO	BBKH, DEKH, QKH, SKH, CCS, VNBA,BBO, GA	Classic 100-Digit Challenge	The performance of the algorithms is ranked as follows: AKQPSO > QKH > CCS > BBKH > DEKH > SKH > VNBA > GA. Subpopulation adjustment has a very clear impact on the precision of AKQPSO Accuracy is very high for classical problems, but it is questionable whether the accuracy of AKQPSO in general problems remains high.	-	[216]
GSA-TLBO	Standard GSA Standard TLBO	Constrained G01 benchmark function	The proposed method has provided the solution to the G01 problem more effectively than other standard algorithms. Test power system / GSA-TLBO / GSA / TLBO IEEE 30 bus system — Fuel cost ($/h): 798.609127 / 798.67514 / 799.0715; Power loss (MW: 8.57558 / 8.386049 / 8.6260 IEEE 57 bus system — Fuel cost ($/h): 41,651.623 / 41,695.8717 / 41,686.7915; Power loss (MW: 15.33112 / N/A / 15.2403 Turkey's 22 bus system — Fuel cost ($/h): 81,170.235 / 82,288.84 / 81,253.99; Power loss (MW: 30.2279 / 43.262 / 36.0993	Economic dispatch problems	[237]
TLBO-CMMI	GA, SFLA, MSFLA, CPLEX GAMS, PSO, GSA, IGSA, DE	PGP considering reliability PGP considering emission Real power system	Performance of proposed technique for test system with a different span of planning considering reliability constraint Span (years) / PGP cost ($\times 10^{10}$ $) / EENS (MWh) / LOLP (Days/year) 12 / 1.2745 / 31480 / 0.0012 14 / 1.5442 / 86.41 / 3.5×10^{-6} 20 / 1.8424 / 469.26 / 1.08×10^{-5} 24 / 2.0469 / 112.05 / 2.12×10^{-6} Performance of proposed technique for 14 years planning year considering emission constraint Test system / PGP cost ($\times 10^{10}$ $) / EENS (MWh) / LOLP (Days/year) / Emission level (%) Without emission constraint / 1.5581 / 188.57 / 7.47×10^{-6} / 13.15 Considering wind system / 1.7219 / 1.64×10^{4} / 4.93×10^{-4} / 9.15 Considering solar system / 1.8626 / 1.03×10^{4} / 3.10×10^{-4} / 8.3 Considering solar and storage system / 1.8996 / 2.67×10^{3} / 9.05×10^{-5} / 8.8 From the result, for the optimum generation mix of PGP solution given by TLBO-CMMI, it is clear that the foremost section of the generation scheme for the Pakistan real power system depends on the local sources and reserves.	Power generation planning	[238]

buses has been determined, and the results show that the integration of PV with BES reduces the system power loss, increases the system capacity, and improves the per-unit system voltage [240]. The authors modeled six stand-alone schemes in Iran consisting of PV-based and wind-based systems considering two forms of energy storage (Hydrogen and battery) to electrify residential loads. The hybrid system consists of electrochemical storage is reliable and more economical than the system with chemical storage. However, in the future, chemical energy storage may be economically reliable with further improvements in fuel cells and electrolyzers. To optimize the designed system based on life cycle cost, a new modified approach, namely the simulated annealing technique based on chaotic and harmony search, has been implemented. The performance of the proposed HCHSA was compared to that of conventional SA and hybrid HS-SA [241]. The authors implemented the SA algorithm to extract four different parameters of the single-diode PV model and solar cell. The proposed SA algorithm follows three steps for parameter extraction: initially without consideration of uncertainty, then, to obtain more precise search intervals, the uncertainty of each parameter has been taken into account, and finally, the parameter value is determined by considering the first two-step results [242]. The authors developed an energy management system and optimized the size of PV and wind power generation by considering three different batteries based on electrochemical technologies. Four recently developed nature-inspired techniques, namely ALO, JAYA, GWO, and KH, have been implemented for this study to assess the lowest electricity cost. The results show that the Jaya algorithm outperforms other techniques. The impact of system lifetime, battery depth of discharge, and battery cost diversity on unit electricity cost has been studied to forecast the system's future performance. It has been found that the hybrid system using a lead-acid battery is economically profitable, followed by the system using a Li-ion battery, and then the next one is the system consisting of a Ni–Cd battery [243]. An efficient HIWO–PSO has been implemented to optimize an integrated off-grid hybrid PV-biomass-battery energy system to feed electrical power to an Egyptian fruit farm. The energy system is designed to meet the required energy for a fruit farm to minimize the total net presence of the system, with consideration of two and three levels of LPSP and EEF, respectively, as system constraints. The prunings from apple and grape trees have been used as a source of biomass, and it was found that about 80% of the power is generated by the biomass system in the optimal PV/biomass energy system [244].

By fusing the merits of individual optimization techniques, the new technique will be able to overcome its shortcomings and attain strong ability between the exploration and exploitation phases. By combining the GA and TS, a hybrid GA-based TS is proposed to determine the grid-connected system components of PV, wind, and biomass-based on the objective of minimizing the system power loss [245]. A hybrid PSO-based HS algorithm has been developed by combined PSO and HS algorithms to obtain off-grid-based system components (PV/wind/fuel) sizes of hybrid renewable energy systems. The system is designed for a location in the northwest region of Iran with the objective of

minimization of net present cost while considering the equivalent loss factor as a design constraint. The optimization results have been compared with the classical PSO and concluded that the proposed hybrid algorithm has attained a high convergence rate with less computational time than PSO [246]. The main drawback of simulated annealing is the requirement of more computational time to determine the optimal solution. By combining SA and TS, a hybrid SA-based TS has been developed to determine the optimal size of a standalone hybrid PV, wind, diesel, biodiesel, FC, and battery-based hybrid system to supply the demand for a region in Greece. In this study, the objective function is based on minimizing the Levelized cost of electricity. The main reason for combining SA and TS is to improve the system quality with less number required simulations. The obtained results have been compared with classical SA and classical TS techniques, and the results prove that the hybrid algorithm is more efficient and flexible [247]. The SA algorithm performs well in obtaining the best solutions in a specific region, and the HS algorithm has a quick convergence time in the region of optimum solutions. The hybrid mixture of the advantages of harmony search and simulated annealing has been combined and developed into a new hybridized HS and SA algorithm that has been proposed to determine the optimal combination of a PV-wind/biodiesel/battery-based on-grid hybrid system for a remote area in Iran that can supply the required demand with a minimum life cycle cost in a reliable way [162]. Many hybrid algorithms have been applied to optimize the size of hybrid renewable systems in recent years. For example, HIS-based SA [41], JLBO by combining the Jaya and TLBO algorithms [248], HGWOSCA [249], SAPSO [250], and others. Hybrid approaches, which are a successful combination of two or more different methods, are used to achieve the best outcome for a given system to overcome the drawbacks of individual artificial intelligence techniques. Although the majority of optimization problems have several objectives, the hybrid approach is thought to be the most effective way to solve them. It has been found various combinations have been applied between SA, CA, TS, PSO, GA, GWO, TLBO, and other algorithms to solve hybrid renewable energy sizing problems, and these combinations seem very effective, able to avoid local optima entrapment and provide superior-quality solutions. Many individual algorithms have excellent exploration capability but suffer from poor exploitation ability, and it has been found that most previous works focused on enhancing the exploitation ability of an algorithm by combining it with other algorithms that have excellent exploitation to improve the balance between the exploitation and exploration phases and enhance their performance. Each algorithm has some novel features for hybridization, and there is a huge interest in combining the features of each algorithm to avoid trapping in local optimality and enhance their capabilities. Over the last few decades, hybrid algorithms have been used to solve several optimization problems with better exploration or exploitation of the search space. It shows greater success in solving different complex problems compared with its counterparts. Thus, when two or more recently evolved precise artificial intelligence techniques are combined, adaptive hybridization opens up a plethora of promising avenues and possibilities for further study. From the results obtained by using hybrid techniques, it has been found that hybrid techniques dominate the output obtained by individual techniques.

6. Recently developed artificial intelligence based optimization algorithms

6.1. Black widow algorithm

Black widow optimization is a new novel population-based artificial intelligence algorithm inspired by the black widow spider's lifecycle. The basics of this algorithm have been shaped according to the unique and bizarre mating behavior of black widow spiderlings. This technique includes three parameters: procreating, cannibalism, and mutation rate, all of which must be adjusted to achieve superior results. The PP rate is the percentage proportion of spiders that reproduce, which provides additional diversification and increases the likelihood of precisely exploring search space. This approach includes three cannibalism stages, and the cannibalism rate omits the improper individuals immediately, which leads to early convergence. Finally, by adjusting the right value of the mutation rate, a fine balance between the stages of exploration and exploitation has been achieved [251]. A novel BWO technique is presented to design the controllers for standalone wind, tidal, and wave hybrid renewable energy systems [252]. The parameters of the PI controller of the wind turbine emulator were optimized using BWOA, and its performance is compared with GA, PSO, and BAT approaches. This technique has not been applied yet to various optimization applications, and to ensure its fitness, it could be necessary to test with a larger iteration [253].

6.2. Sailfish optimizer

Sailfish optimizer is an optimization algorithm that has been inspired by the group hunting behavior of sailfish. Initially, the SFO method imitates the populations of predators (sailfish) for intensification and prey (sardines) for diversification and then employs a complex form of attack-alternation hunting strategies on sardine prey to break down its collective defense. Finally, this method updates prey movements over the search space, and predators are allowed to catch the prey and relocate its past position to the new fittest. Sailfishes are assumed to be candidate solutions, and problem variables are defined as sailfish position. The best position of the fittest sailfish and the injured sardine are saved in each iteration, and the highest fitness position at the iteration can have a dramatic impact on SFO performance. SFO has several advantages: the ability to avoid local optima, fast convergence on large-scale global optimization, and competitive results for an improvement in the balance between exploration and exploitation phases [254]. To improve BSF exploitation ability, a hybrid adaptive-hill climbing with Binary Sailfish optimizer method is proposed and applied to 18 standard UCI datasets; the results obtained are compared with ten artificial intelligence feature selection approaches [255].

6.3. Deer hunting algorithm

The deer hunting algorithm is a human-based artificial intelligence optimization algorithm that imitates the hunting behaviors of humans towards deer or bucks based on considering wind angle, the position of deer, and so on. The DHA begins with the vector that represents the number of hunters in the population and initially assumes the position of leader, with succeeding hunters close to the optimum. The position of the hunter will be continuously updated until it achieves the best position based on the coefficient vectors. The position angle can then be updated based on the angle difference between the wind angle and the prey visualization angle, and finally, the position of the successor is used to update, allowing for global search. Its efficiency was effectively proven over 39 benchmark problems and 3 constrained engineering applications compared with other state-of-the-art optimization approaches. The main problem with DHA is its low premature convergence due to a large number of random parameters [32]. An optimal combined solar chimney power plant with SOEC, SOFC, and an HRSG was presented for supplying power to residential consumers in Yazd, Iran. An improved deer hunting technique is utilized to analyze the economic aspects of the combined solar chimney system. The outcomes of the presented method were compared with GA and PSO [33].

6.4. Tunicate swarm algorithm

The tunicate swarm algorithm is a bio-inspired artificial intelligence optimizer in which optimal sources are derived from the swarming behaviors of marine tunicates and their fluid jet propulsion. Tunicates are typically a few millimeters in size, with each tunicate having a common

gelatinous tunic and special abilities to produce pale blue-green light, allowing it to see food sources thousands of meters away in the deep ocean. A tunicate must satisfy three conditions to mathematically model jet propulsion behavior: (a) avoid conflicts with other search agents and, taking gravity and social forces into account, introduce a vector to determine new search agent position, (b) move search agents towards the position of the best neighbor, and (c) converge close to the position of the best search agent. The first two global solutions of search agents, on the other hand, are saved to mathematically simulate swarm behavior, and other search agents update their position toward the best optimal solution. A new improved tunicate swarm algorithm is introduced for concurrent optimal allocation and control of the CBs, DGs, and DNR to minimize distribution network power losses while providing quantitative and qualitative power service to meet consumer requirements [256]. The proposed ITSA is applied, tested, and compared on standard 33-bus radial distributed systems, 69-bus systems, and large-scale 119-bus radial distribution systems, considering different rated power demands and their operating strategies. Also, the incorporated ITSA results are compared with other approaches such as conventional TSA, PSO, GA, EA, WCA, BFOA, and CSA [257].

6.5. Artificial electric field algorithm

The artificial electric field algorithm works on the principles of Coulomb's law of electrostatic force. In AEFA, the agents are the charged particles, and only the best charge particle (with the greatest charge) moves slowly in the search space by considering the attraction of electrostatic force. It is defined as a memory-based one, and the position and velocity of each search agent are updated by a particle's personal best and global best fitness history. Its inspiration and performance have been effectively proven over 51 benchmark problems on learning-based real-parameter single-objective optimization compared with other existing state-of-the-art optimization approaches. Also, theoretical analysis has been investigated to confirm the stability conditions for the convergence of particles [258].

6.6. Water strider algorithm

The water Strider algorithm is based on the life cycle of water strider bugs, which mimics their territorial behavior, mating process, ripple communication system, foraging behavior, and succession. WSA commences the search process by determining the initial position of each water spider on the lake, and WSs are evaluated using an objective function to find their fitness. Then establish several territories and assign the positions of WSs in each territory based on their fitness. The new position of the male keystone is updated based on the response signal from the female for the mating signal. Keystone is typically more energy-consuming during the mating process, and to compensate, it requires a new position to find the availability of its food. The objective function is evaluated to determine the location of food availability. If you aren't able to find food, Keystone will die of starvation. In this case, the killed keystone is replaced by a new, mature larva, and its position is randomly initialized inside the territory. The feature of this algorithm was tested by several numerical and engineering design problems, and the results have been reported as providing better performance than compared with other state-of-the-art algorithms [259].

6.7. Political optimizer

The Political Optimizer is a new socio-inspired optimization algorithm that is inspired by the process of multi-phased politics. PO mathematically maps the sequence of five different phases of the political process such as "constituency allocation, party formation, election campaign, inter-party election, and parliamentary affairs". In the phases of party formation and constituency allocation, the total population is divided into several political parties, where each party consists of several members, and several constituencies are also assumed. Both of these phases execute just once, for initialization only. In the election campaign phase, a novel method of updating positions is mathematically formulated with the aspects of learning from the previous election and updating the positions of members and candidates concerning the party leader and constituency winner, respectively, to help improve the performance of the candidate in the election. In the inter-party election phase, the election is simulated in each constituency, and the winning candidate is announced. In the parliamentary affairs phase, the winning candidates from all the parties then collaborate to form and run the government. The entire population is divided into many political parties, each of which has several candidates or members, and the learning behavior of politicians from the previous election is mapped by a recent past-based position updating strategy, and the candidates' performance has improved. By instilling the concept of logically dividing the population into political parties and constituencies, PO assigns a dual role to each individual (Party member or candidate) solution. Instead of the best solution, PO interacts with two improved solutions, such as the best leader of the party and the best winner of the constituency, to update the search agent's position. The performance of the developed algorithm has been evaluated on various optimization problems against 15 well-known state-of-the-art algorithms [260] Optimization techniques have been rapidly growing during the past few decades. However, as the complexity of the problem increases, the probability of achieving a global solution decreases. Recently, several optimization techniques have been proposed, which include the Arithmetic Optimization Algorithm inspired by the distribution behavior of mathematical operators [261], the Slime mould algorithm inspired by the food searching behavior of Physarum Polycephalum [262], the Sparrow search algorithm based on group wisdom, foraging and anti-predation behaviors of sparrows [263], Orca Optimization algorithm inspired by Orca's hunting approach [264], Strawberry Plant Propagation Algorithm, etc. [265].

7. Software tools applied to optimal sizing of HRES

Before initiating the proposal, it is critical to model the energy infrastructure for decision-making and pre-feasibility review. Several commercial and free software resources are available for analyzing the integration of renewable systems. These include ARES, CREST, iHOGA INSEL, HOMER, HYBRID 2, PHOTO, PVF-Chart, PV-DESIGN-PRO, PVSYS, RAPSIM, RAPSYS, PVSYST, RETScreen, REopt, SOLSIM, SOMES, TRNSYS, WATSUN-PV. RETScreen and HOMER are the most popular applications among academics, with over 200,000 and 28,000 installations, respectively [266]. The authors carried out a comprehensive review of the environmental analysis of HOMER, iHOGA, EnergyPRO, RETScreen Expert, and TRNSYS. This analysis aimed to gain an understanding of the methods used to develop community-level energy simulation software for quantifying system environmental efficiency [267]. Using the simulation tool TRNSYS, the performance simulation of a 200 KW solar power system at VIT University has been examined [268]. The authors modeled a hybrid geothermal-solar-wind system and performed energy, technical, and economic analysis using TRNSYS software, which supplies homes with space heating and cooling, indoor hot water, and electrical resources [269]. Also, TRNSYS tools are utilized to assess the effect of advanced control techniques on the energy and economic efficiency of a residential heat pump device equipped with a photovoltaic field and electrical energy storage that runs in a broad variety of temperature conditions [270]. A hybrid PV/WT/DG/Battery energy system has been optimized for supplying the electrical load of a residential house in Jahazpur, India. For this HOMER to perform technical and economic analysis, and then used RETScreen to study financial viability, payback period, risk analysis, and the effect of taxes and subsidies for an optimized configuration [271]. Using the RETScreen software tool, the optimal sizing of renewable energy systems for the different objective functions has been determined for two

problem descriptions, with and without considering subsidies for renewable energy systems. [272].

Single and multi-objective optimization for a PV/WT/DG/Battery HES for a typical house in France was performed using iHOGA. In comparison to the single goal of minimizing the NPC, multiobjective optimization resulted in significant reductions in CO_2 emissions and unmet load while maintaining a slightly higher NPC. The iHOGA tool has been used to optimize an isolated hybrid PV/DG/Battery for the inhabitants of Mucura island and the results prove that it is feasible to minimize a microgrid's operation costs by reducing the reliance on diesel fuel, optimizing the usage of local renewable energy sources, and ensuring energy availability across the year [297]. By using the HOMER simulation tool, a techno-economic analysis has been performed for a renewable power system comprising PV/WT/DG to electrify remote areas [298]. To meet the required load demand for six different climatic locations in Tamil Nadu, India, the author designed an off-grid hybrid renewable energy system using HOMER and discussed various feasible combinations that can be economically and environmentally suitable for each location. Also, sensitivity analyses have been performed for six different parameters to examine their effect on system performance [299]. To simultaneously meet the electrical, thermal, and hydrogen load demands, an off-grid or on-grid hybrid renewable energy system containing solar, biomass, and hydro has been designed using HOMER to meet the requirements of the tea industry. The results show that an on-grid system outperforms an off-grid system in terms of the Levelized cost of energy, Levelized cost of hydrogen, and emissions [300]. The issues regarding integrating hybrid renewable energy systems into smart microgrids have been discussed [301], and the challenges associated with designing hybrid renewable energy systems have been discussed, and a feasibility analysis of six different configurations has been performed using HOMER for remote areas located in Madhya Pradesh, India. It has been found that the hybrid system consisting of solar, wind, biomass, diesel, and batteries is more feasible than other combinations [302]. The summary of studies that applied HOMER software to optimize HRES is presented in Table 8.

Not all software packages can accomplish all the tasks conceivable. However, the choice of specific tools depends heavily on scenario objectives. There are many separate platforms available for conducting detailed assessments of the effectiveness of energy sources, conducting feasibility studies for electricity generation, and studying thermal and electrical systems. HOMER has been used the most widely and continuously because HOMER has already connected all renewable energy resources, as well as each renewable energy resource's technological component. However, it excludes solar thermal systems. RETScreen is a widely popular technological platform for assessing the feasibility of renewable energy technologies. RETScreen was developed by the Government of Canada and the first version was released on April 30, 1998. It has been widely used as a research tool by researchers to design clean energy projects. This tool has five different stages of a process, which include an energy model, cost analysis, GHG analysis, financial summary, sensitivity analysis, and risk analysis. The current version of this software is RETscreen Expert 9, an integrated software platform that was released on October 13, 2022. Unlike the previous version, version 9 includes notable features such as benchmark analysis, feasibility analysis, performance analysis, and a portfolio analysis module. This tool has been updated for 64-bit environments, which improves memory usage and performance, and it is available in 38 languages, the most recent of which is Albanian. It requires less skill to use, and the requirement of fewer input parameters and less computing power is two of the main advantages of this tool. RETScreen has a facility for combining power, heat, and cooling purposes. It can only analyze solar, wind, and battery energy and cannot analyze generator, hydroelectric, bioelectric, or thermal energy capabilities, and time series data files cannot be imported [303]. On the other hand, no sub-tool or function is provided for optimizing submodels of interconnected energy systems. iHOGA PRO + also has features like pre-sizing, PV slope optimization, probability analysis, battery aging modeling, and multi-objective optimization, but it has only been used in a few studies so far. The primary drawback identified with iHOGA PRO + is the absence of information on the surplus energy available from any of the energy technologies consideredThe techno-economic analysis is possible with HOMER and iHOGA software; but, these tools do not perform thermal analysis, whereas RETScreen and TRNSYS platforms perform both techno-economic and thermal analysis. All software tools can not optimize all renewable sources. TRNSYS and iHOGA, for example, cannot optimize bio-energy systems, and RETScreen and TRNSYS do not have features to optimize hydro energy.

There are several software tools available for analysing the performance, unit size, power supply losses, and energy production cost of various hybrid system configurations. Battery banks are the only type of electrical energy storage that most software programs can model; the only programs that can model hydrogen energy storage are HOMER and iHOGA. HOMER is the most powerful tool to optimize the designed hybrid system, which combines engineering and economy, allows the users to determine the least cost value quickly and effectively, displays numerous choices of optimized designs based on the least net present cost, easily understood, and displays their results in an efficiency graph. The demerits of this software are it can able to apply only the first degree of linear equations and is not possible to import time series of metrological and load data [304]. Secondly, the only simulation tool that can integrate compressed air energy storage is HOMER [305]. In contrast, HOMER only takes into account net present cost as a single objective function and ignores intra-hour fluctuation. Also, HOMER does not rank the system based on the cost of electricity and ignores the bus voltage variations. It is also important to note that HOMER does not account for the impact of a potential future load increase or the discharge depth of batteries. Due to the inversely proportionate relationship of depth of discharge with battery capacity and life span, the depth of discharge is essential to HRES optimization. For better optimization outcomes, this area of the HOMER software has to be improved [13]. iHOGA is one of the versions of the Hybrid Optimization by Genetic Algorithms software, developed by researchers at the University of Zaragoza for the simulation and optimization of renewable energy-based microgrid energy systems. It includes multiobjective optimization and can provide an optimal solution using very low simulation step times. Their main disadvantage is that the daily load is limited to 10 kW. A simulation tool called TRNSYS was created at the University of Wisconsin. Both passive and active solar design, it is largely employed in the disciplines of renewable energy engineering and building simulation. It offers the choice of incorporating models from their library, as well as digital simulations and accurate unit sizing. Its ability to simulate transitory integrated system behavior is one of its key merits [87]. The most popular software programs applied for sizing renewable energy systems, according to the literature, are HOMER and RETScreen. Some advancement are necessary to be made in software tools such as the integration of alternative energy sources and electric vehicles, operating management, the inclusion of various social and environmental indicators, and the cost of emissions which will be useful in future studies and hybrid system applications.

8. Discussion and findings

This work focused on classical approaches, artificial intelligence based optimization techniques, hybrid algorithms, and commercial software tools for the techno-economic analysis of hybrid renewable energy systems. Despite having great performance, the ability to tackle complex issues, and the ability to use many objective functions, the main disadvantage of most of the optimization techniques are complex and case sensitive nature. Classical techniques are the most useful strategy in economic optimization, but suffer from limited space in which to optimize parameters. Classical methods have several drawbacks, including rigidity, difficulty in locating a global optimum, and difficulty

Table 8
Summary of studies that applied HOMER software to optimize HRES [273–296].

Table 8.
Summary of studies that applied HOMER software to optimize HRES

System components	SA/GC	Analysis	Location/Application	Objective function	Sensitive Parameters	Findings	Ref.
PV/WT/DG/BT	SA	Economic-Feasibility, Sensitive	Adafoah,Ghana / Rural areas	To determine to optimize system based on Minimum NPC and COE	Solar radiation Wind speed Diesel price	Optimal system NPC($) COE($/KWh) PV/WT/DG/SB 3905.6 0.281 The impact of PV cost on the LCOE is insignificant in the regions of Ghana	[273]
PV/WT/BT	SA	Techno-economic	Kenya and South Africa / Nitric acid plant	Minimization of TNPC	-	PV/WT/SB (Optimal system) Kenya South Africa Scenario A COE($/KWh) 0.200 0.235 Scenario B COE($/KWh) 0.163 0.079	[274]
PV/DG/BT	SA	Economic Environmental	12 Different locations in Nigeria / Transceiver Station	To Minimize LCOE, CO_2 emissions	-	Optimal system COE($/KWh) Solar hybrid systems Ranges between 0.156 to 0.172	[275]
PV/FC/HT/BT	SA	Techno-economic Environmental	NEOM,Saudi Arabia / Seawater desalination plant	To minimize NPC, COE	-	Optimal system NPC($) COE($/KWh) PV/FC/SB 438,657 0.117 The PV/FC/BS system is better than the utility grid extension up to a distance of 16.1 km.	[276]
PV/WT/BG//PHS	SA	Techno-economic	Djounde, Cameroon \ Remote village	To minimize NPC, COE	Wind speed Solar radiation Capital cost multiplier of PV and PHS Biomass price and availability Maximum capacity shortage	Optimal system NPC($) COE($/KWh) PV/Biogas/PHS 370426 0.256 Beyond the distance of 12.78 Km, the PV/BS/PHS off-grid system is considered a better solution for Djounde's electrification than grid extension.	[277]
PV/WT/DG/NG	SA	Economic Environmental Reliability	Medha, Oman / Exclave territory	To obtain Low NPC, COE To reduce GHG emissions	-	Optimal system NPC($) COE($/KWh) WT/NG/DG 48.9M 0.0761	[278]
PV/HYDRO/BG/DG	SA	Techno-economical Environmental Sensitive	Melky Hera, Ethiopia	To Minimize NPC To increase RF To reduce pollutants	Solar radiation Diesel fuel price	Optimal system NPC($) COE($/KWh) PV/Hydro/DG/SB 997334 0.133 The obtained $0.133/kWh COE is greater cost competitive than the present current grid price of $0.06/kWh, but somehow good	[279]
PV/DG/BS	SA	Economic Sensitive Environmental	Khorfakkan, Sharjah	To analyze NOC, COE for single-axis and dual-axis PV system To obtain the lowest GHG emission	Derating factor Diesel price	Optimal system NPC($) COE($/KWh) PV//DG/SB (Dual axis system) 44,483,730 0.25 Sensitive analysis reveals that as the derating factor and diesel fuel cost decrease, so does the LCOE.	[280]
PV/BM	GC	Techno-Economic Sensitive	Tripura, India / Rice mill	To improve RF To decline the NPC, COE	Solar radiation Variation in Load Electricity rate Annual capacity shortage	Optimal system NPC($) COE($/KWh) PV/BM 78,980 0.143 The proposed hybrid system may conserve over 90% of utilization from the grid for rice mills.	[281]
PV/WT/BG/NG/DG/BT	SA	Techno-Economic Sensitive Emissions	Norway / Household applications	To Minimize NPC, COE, To decrease the environmental impacts	Prices of diesel, natural, and Bio-producer gas	Optimal system NPC($) COE($/KWh) PV/WT/Bio/SB/Capacitor 72,232 0.306 Wood gas generators served most of the electricity demand followed by the wind turbine and the PV array	[282]
PV/DG/BT	SA	Techno-Economic Emission	Manitoba, Canada / Remote area	To determine the optimal system configuration	-	Optimal system NPC(CAD) COE(CAD/KWh) PV/DG/SB 21,364,520 0.5158 Both solar and Li-ion batteries are considered environmentally safe and pose no risks of greenhouse gas emissions.	[283]
PV/WT/BT	GC	Adaptability Economic Environment	Warrnambool, Australia / University building	To determine optimal design with the least NPC and COE	-	Optimal system NPC($) COE($/KWh) PV/DG/SB 5.68M 0.0851	[284]
PV/WT/DG/SB	SA	Techno-Economic Environmental	Barwani,India / Remote Village	To minimize cost functions To reduce Pollutant emissions	Solar and wind resources Diesel price	Optimal system NPC($) COE($/KWh) PV/WT/SB/DG 3,24,178 0.6213	[285]

PV/WT/BG/SB	SA/GC	Techno-Economic Sensitivity	Leopard Beach, China / Remote Village	To minimize system NPC, COE	Project lifetime Nominal discount rate Annual average electric loads	Optimal system NPC($) COE($/KWh) PV/WT/BDG/SB 547013 0.201 The proposed PV/Wind/BDG/Battery hybrid system is more economical than the grid extension, in which the Breakeven Gird Extension Distance is 9.71 km.	[286]
PV/FC/SB	GC	Techno-Economic, Sensitivity	Cheras , Malasiya / Medical CentreBuilding	To analyze the effectiveness of economical parameters and the impact of sensitive variables	Fuel prices Annual interest rate	Optimal system NPC($) COE($/KWh) PV/FC/Battery 106551 0.091 Sensitivity analysis revealed, with the increase in annual interest rate from 1% to 6%, the TNPC decreases whereas the LCOE increases	[287]
PV/WT/FC/SB	SA	Techno-economic	Nikouyeh Ghadamgah Moaleman Marvdasht, Iran / Research center	Minimize total energy cost	-	Moaleman Nikouyeh Ghadamgah Marvdasht PV/WT/SB (Optimal system) COE($/KWh) 0.54 0.56 0.799 0.62 Battery storage is more economical than fuel storage.	[288]
PV/DG/SB	SA/GC	Techno-Economic Sensitive	Mandhra Saidan, Pakistan	Minimize NPC and LCOE	Solar PV price Diesel fuel price Battery cost Exchange rate Inflation rate Discount rate	NPC($) COE($/KWh) PV/DG/SB (Optimal system) Offgrid 1,444,310 0.145 PV/DG/SB and grid (Optimal system) On-grid 1,042,344 0.072	[289]
PV/WT/BG/DG	SA	Technical Economical	Newcastle, United Kingdom / Residential	To calculate and compare NPC and COE of different system configurations	Number of components	Optimal system NPC($) COE($/KWh) WT (1KW) /BG/SB 14507 0.588	[290]
PV/WT/SB	SA	Techno-Economic Sensitive	Medicine Hat, Canada / Remote oil well	To determine optimal design in the order of least NPC, LCOE, and operating cost	Solar radiation Wind speed	Optimal system NPC($) COE($/KWh) PV/WT/BS 145,150.50 0.5118	[291]
PV/Hydro/DG/SB	SA	Techno-Economic Environmental Sensitive	Sakran, Traq / Rural village	To examine optimal configuration based on LCC, COE, and GHG emissions	Pipe head losses Generator under loading Battery roundtrip efficiency State of charge Capacity shortage PV capital cost Multiyear inputs	Optimal system NPC($) COE($/KWh) PV/battery/hydro/diesel 145,150.50 0.054 The NPC for the multi-year module during the project's lifetime of 20 years shows an increment of 22.5% in comparison with that of the single-year module	[292]
PV/WT/FC/HT/EL/SB	SA/GC	Techno-Economic	Minya Governate, Egypt / Water pumping application in a remote area	To rank the optimal design based on NPC and COE	-	Optimal system NPC($) COE($/KWh) PV/FC 50321 0.058 All standalone HRES systems are more feasible and economical than grid extension except stand-alone diesel generator systems and wind/ battery systems.	[293]
PV/WT/Hydro/DG/SB	GC	Techno-Economic Sensitive Environmental	Fouay, Benin / Rural electrification	To find optimal configuration based on NPC, COE, and GHG emissions	Diesel fuel price Load growth Interest rate Annual average stream flow Project lifetime	Optimal system NPC($) COE($/KWh) PV/DG/SB (Off-grid) 0.207 555492 The Off-grid PV/DG/SB system is more cost-effective and economical than the grid electricity tariff (0.22 ($/KWh)	[294]
PV/SB	GC	Techno-Economic Sensitive Environmental	Jalalabad, India / Electrification to Rural village	To obtain minimum LCOE, TNPC with maximum RF To ensure minimum CO_2 emission	Cost of Grid energy	Optimal system LCOE (With DR and ORM) SPV, BESS, and grid 11.23/kWh (Grid tariff Rs.10/kWh) 8.36/kWh (Grid tariff Rs.6/kWh) COE produced in BESS coupled with the grid is more economical than the grid with the SPV system.	[295]
PV/WT/BM	GC	Techno-Economic Sensitive Environmental	Kallar Kahar, Pakistan	To determine minimum system cost and COE	Wind turbine hub height PV Capital cost multiplier Solar radiation Biomass price	Optimal system TSC($) COE($/KWh) PV/BM/WT/Grid 180.2 M 0.05744 Grid-connected hybrid systems are more effective than off-grid hybrid systems.	[296]

responding to dynamic changes. Artificial intelligence algorithms overcome the deficiencies of classical methods and provide a globally optimal solution with a fast convergence speed. Algorithm results have been compared in terms of technical, economic, and performance. It has been observed that artificial intelligence algorithms provide better performance in solving multiobjective optimization problems. Artificial intelligence techniques are the most employed in optimization due to their higher efficiency, greater accuracy, and fast convergence rate; however, they are more complex and require a perfect balance between the exploration and exploitation phases. Due to the complicated processes and codes employed, the artificial technique requires significant hardware performance to function. However, these methods have the advantage of being fast and accurate while still being highly efficient. Also, hybridizing two or more algorithms can help solve problems more quickly, reliably, and effectively. The combination of various artificial intelligence techniques demonstrates a hybrid approach to system design that is distinguished by high speed, rapid convergence, and robustness but requires more complexity to design the system and suffers from code difficulty. Commercial software tools are easy to use and versatile, but their application to multiple objectives and unsophisticated optimization is limited. While comparing commercial software and classical techniques, software-based tools are easier to employ, offer more flexibility, and require less time. But most of the software-based tools are black boxes and limited by the application of a simple optimization equation. Several problems, such as finding the exact location to install hybrid renewable energy sources and the potential of renewables (solar, wind, biomass, and hydro), are difficult, especially in remote locations. Hence, exact geographic software tools are necessary to develop.

Most of the work considered single objectives, which could not produce quality outcomes, so to attain realistic outcomes, multiple objectives, which include social, environmental, and uncertainty factors, needed to be considered simultaneously. A massive number of studies have been conducted to develop hybrid renewable energy systems that can meet the electrical energy demand alone. The development of combined heat and power, hydrogen production, and hydrogen energy storage have a great potential to fulfill their requirements in multiple sectors for multipurpose applications, which would be an attractive research direction. As global warming is the major concern in most of the studies, the generations of thermal energy and hydrogen production from renewable energy sources have not been considered. Furthermore, there is no one software program available to create a hybrid energy system that uses hydrogen energy technology. The performance of the hybrid renewable system has not been analyzed through sensitivity analysis based on load variation, weather conditions, multiyear life of the project, interest rate, and component degradation time. The consideration of social aspects such as human progress and providing employment opportunities should be considered major challenges while optimizing hybrid renewable energy systems. The high capital cost to install a hybrid renewable energy system leads to a prolonged payback time, and several technical improvements have been required to further reduce the manufacturing cost of renewable components, which drives down payback time. Many new algorithms have been developed that can be used to achieve better results than standard algorithms. More efforts are needed to investigate and create hybrid optimization methods to compete with the existing problems of optimizing hybrid systems.

As previous research has shown, it is clear that implementing an efficient HRE design presents several challenges. The following

summarizes some of the findings.

- The use of classical size optimization methods has been declining in recent years, while an increasing shift toward the use of modern size optimization methods has been observed, as modern techniques provides a range of optimal outcomes that allow decision-makers to choose the most appropriate combination of HRES. There are significant differences were observed in system components and electricity costs generated by a single algorithm, hybrid algorithm, and software tools. As a result, it is proposed that modern optimization approaches be used in HRES because they can offer promising and practical optimum sizes. To improve the performance of the optimization algorithms it is important to manage the balance between diversification and intensification.
- Recently, several new Artificial intelligence algorithms have been developed based on various inspirations but have not been widely used for solving complex optimization problems related to hybrid renewable energy systems. As can be seen from this review, newly developed hybrid algorithms such as CSA, Jaya, and DHO provide better-quality results with less computational time, better reliability, and accuracy compared with GA and PSO. Many scholars, especially in the field of optimization, have expressed interest in the PPO, TSA, GEO, Bayesian, and Resilience optimization algorithms. Implementing the above-mentioned algorithm, there is no written research in the field of green energies. Hence, recently developed algorithms have to be employed in the field of optimal sizing of hybrid renewable energy systems, and their performance has to be assessed.
- Concurrent integration of renewable sources and handling of constraints may significantly add to the complexity of the model, depending on the multiple objective functions. Therefore, a convenient optimization approach with better convergence, an optimum solution, a high level of accuracy, and fewer control variables is necessary to manage such difficulty effectively.
- Essentially, software tools offer the most economically viable approach and do not direct the consumer to a predefined style of solution dependent on performance parameters. The fundamental optimization algorithm in software tools is a “black box” that is shrouded by the consumer and does not provide details about whether or how the outcomes are achieved.
- The desired technical performance and the dynamic response of the power system depend upon the optimal sizing of renewable energy components. There is a scarcity of tools for designing and optimizing multiple energy systems in different consumption sectors. Hence, an integrated tool to address multiple energy resources, different load sectors, and properties to analyze the effect of components and resource uncertainty has to be designed.
- Due to the nonlinearity of the diesel generator characteristic, further research into other forms of renewable energy sources such as ocean energy (wave and tidal) and bio-fuel from seaweed, as well as the most promising storage technology is expected, which promises to reduce costs and make remote sites become viable options for renewable energy systems.
- Since hydrogen is a potential option for energy transfer, its major issue is its high production cost, and storing the hydrogen efficiently and safely is a challenging one. Hence, further research and effort are needed to make hydrogen available for public use in future hydrogen-based energy systems.
- The recycling of storage batteries is a significant problem for both consumers and suppliers. However, recycling and reusing these batteries would maximize the system’s beneficial ecological effects. Much more studies need to be conducted in the future to improve the efficiency of batteries. Many reports only considered the environmental implications of energy storage and diesel power generators and overlooked the environmental impacts of pumped hydro storage facilities and their land requirements.
- For the optimal design of a hybrid renewable energy system, hourly load forecasting plays a significant role, and hence the estimation of changes in load behavior for future generations is required. An accurate prediction model analysis is required to estimate more accurate estimates of load profile.
- Climate change has a significant impact on the robustness of hybrid renewable energy system. Because the local climate has a large impact on energy demand and resource supply, the consequences of changing weather conditions should be considered in future research to determine whether the robustness of hybrid power systems under climate change has been enhanced or not.
- Feed-in tariffs, tax credits, and other incentives implemented by state and federal governments all have the potential to significantly increase HRES consumption.
- Carbon costs must be levied on all types of power stations in proportion to their carbon emissions.
- Investment and operation-related costs of all systems vary considerably across studies; therefore, the total present cost and cost of energy outcomes can be substantially different. Based on statistical approaches, specific regional cost factors may be described to correct the cost disparity.
- Only a few studies have taken into account technical, economic, social, environmental, and political criteria when examining HRES-appropriate configurations. The environmental impact of renewable energy components, land requirements, and their associated costs, as and social impacts such as energy poverty reduction, health impact, employment opportunities, etc., have been taken into consideration while optimizing hybrid renewable energy systems.
- The fossil fuel consumption of sea and air transportation significantly emits CO_2 and NO_X into the atmosphere. It has been observed that the integration of renewable sources has been applied to applications such as residential, industrial, and commercial; however, it is limited to sea and air transportation. As a result, alternative, non-zero fuels are required in a variety of transportation sectors. The use of new energy sources in ship and airway power systems is insufficient, and research on ship and airway-based energy systems are likely to be the subject of research in the immediate future.

9. Conclusion

This article aims to analyze the optimization methods applied to optimize hybrid energy systems, including objective functions and different evaluation criteria, to gain a deeper understanding of how to solve the optimization problem. Among various software tools, HOMER is the most widely used software tool as it allows for the maximum combination of renewable energy and conducts optimization and sensitivity analyses that make it simpler and quicker to evaluate several possible system configurations, but it is limited by using a single objective function. Nowadays, researchers are focused on artificial intelligence techniques, which are characterized by being deemed more acceptable than conventional approaches due to their ability to look for global optimal, high calculation precision, and rapid convergence speeds, but artificial techniques still face challenges due to the greater number of required inputs. Hence, there is a need for much more precise and sophisticated optimization techniques. It has been identified that hybrid algorithms have attracted much more attention than single algorithms, which can provide more accurate results and have the ability to solve multiobjective optimization problems effectively. In the future, researchers should focus on rigorous research to develop effective artificial intelligence techniques, hybridize them, and handle various objective functions effectively.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

Data will be made available on request.

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